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(54) **SINUS-ENGAGING VALVE FIXATION MEMBER**

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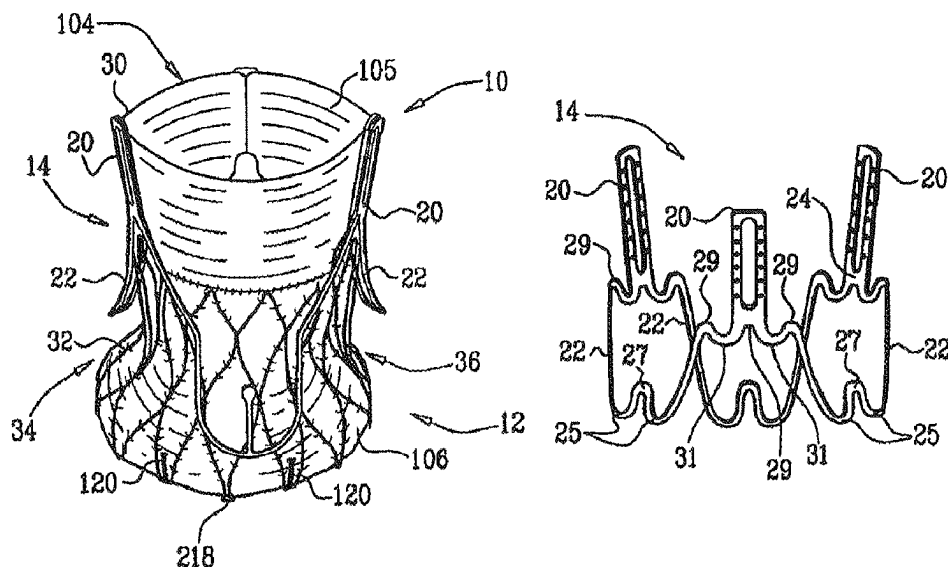
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(57) **ABSTRACT**

A method of delivering and deploying a valve prosthesis
includes delivering the valve prosthesis disposed in a deliv-
ery catheter to a native valve, proximally retracting a deliv-
ery catheter tube of the delivery catheter such that three
engagement arms of a support of the valve prosthesis are
released from the delivery catheter tube and flare outwardly
while an inflow frame portion of the support remains within
a distal portion of the delivery catheter, and distally advanc-
ing the distal portion of the delivery catheter such that the
inflow frame portion is released from the distal portion.

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FIG. 1

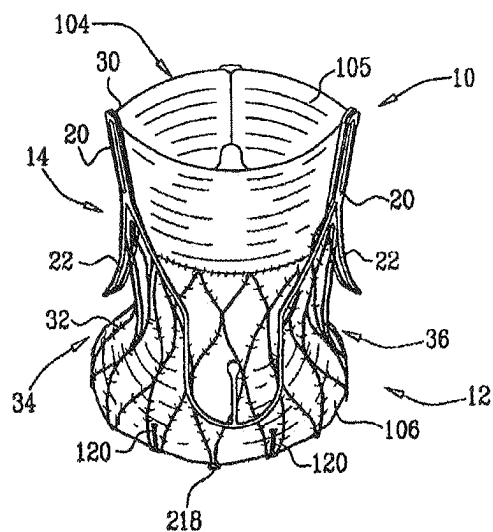


FIG. 2A

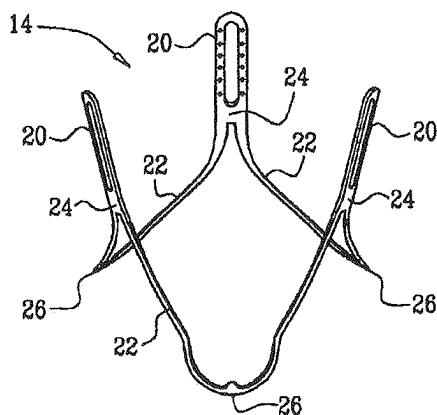
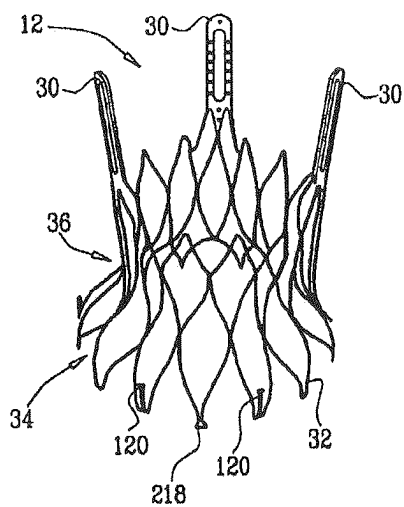


FIG. 2B



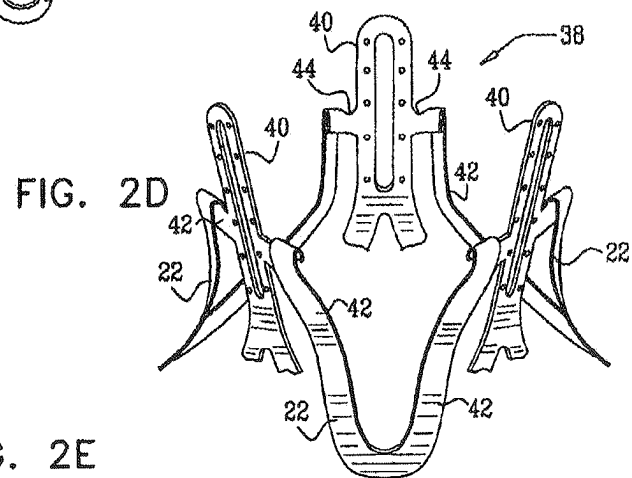
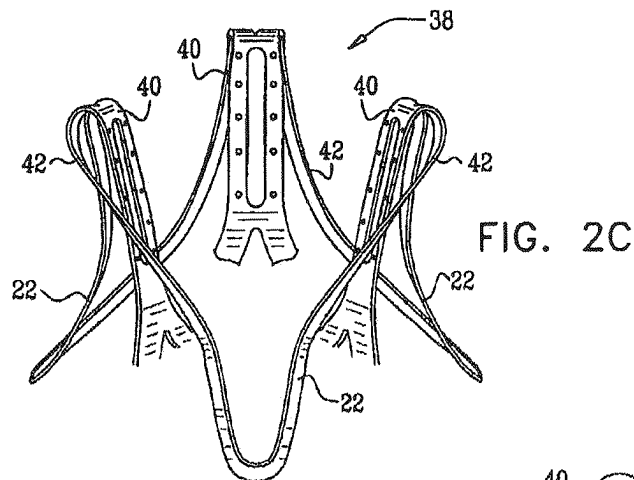


FIG. 2E

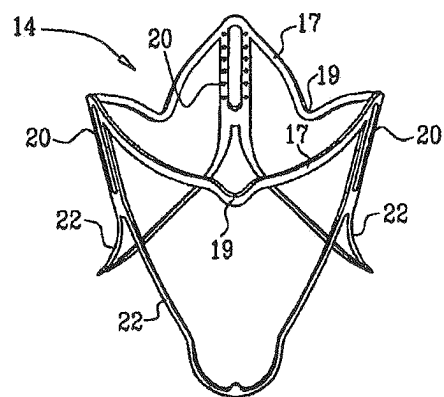


FIG. 3A

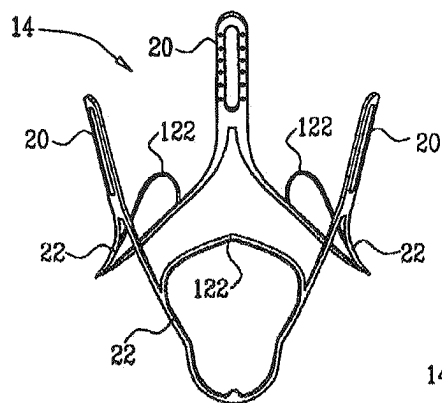


FIG. 3B

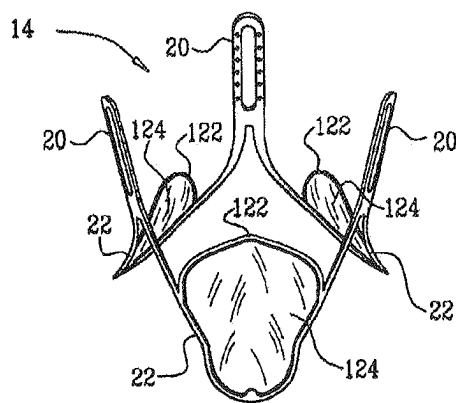


FIG. 3C

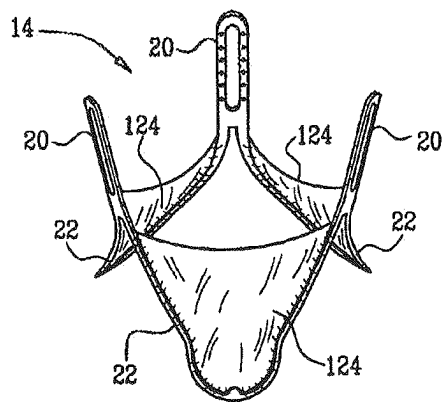
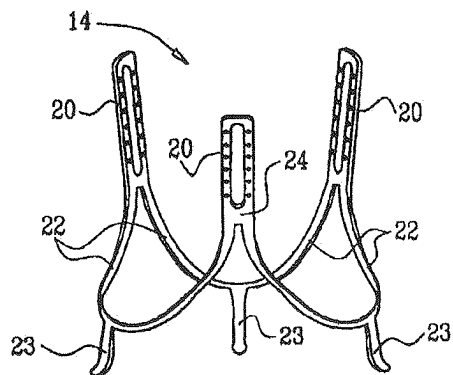
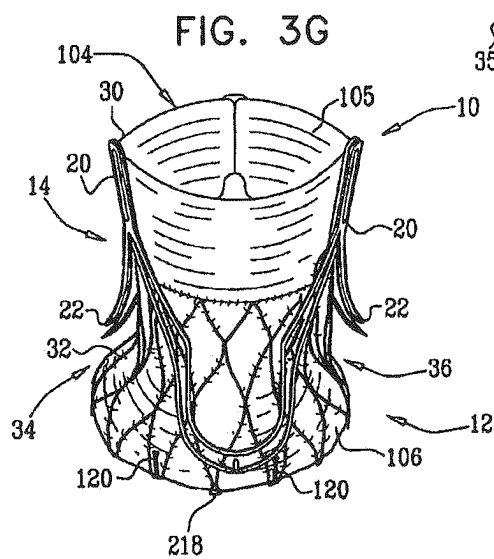
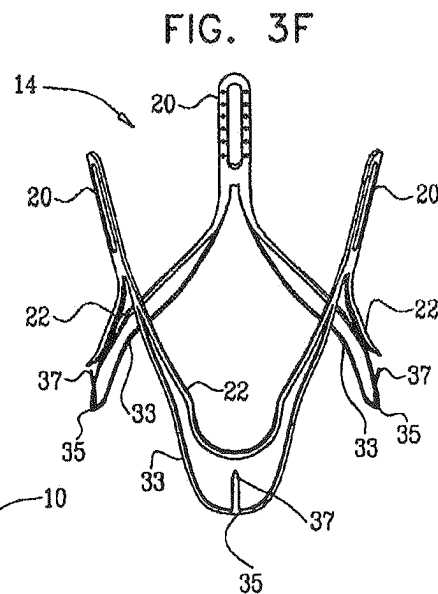
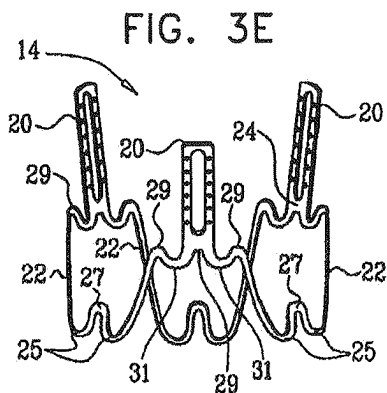
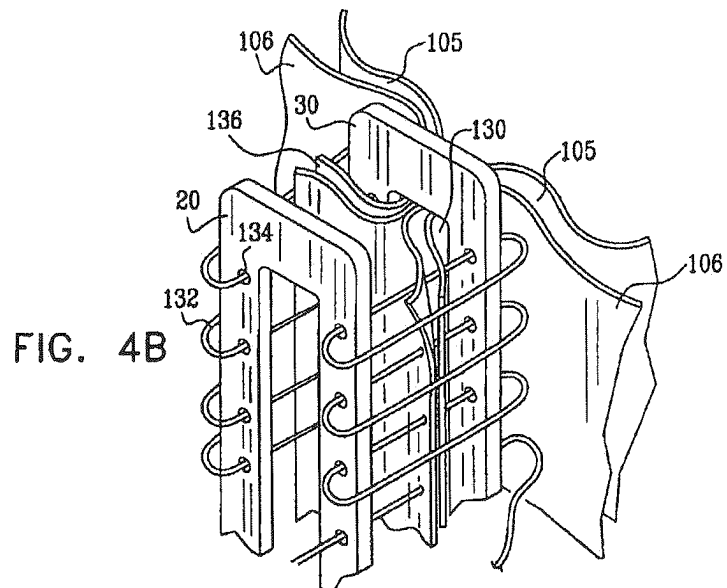
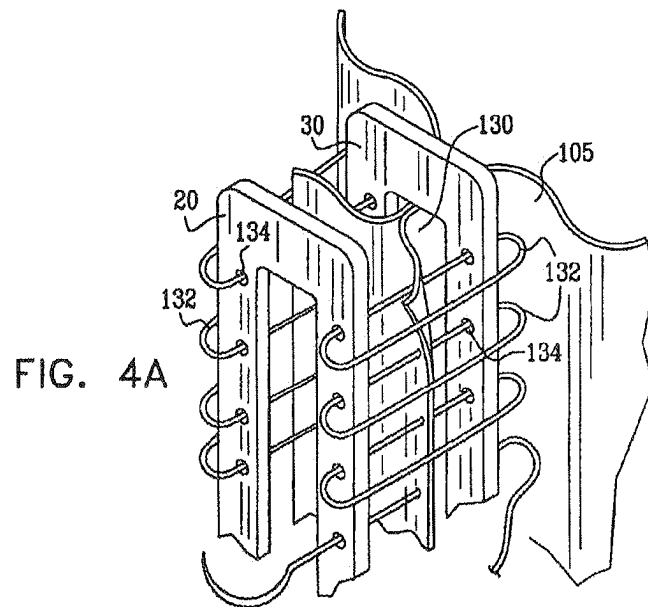


FIG. 3D







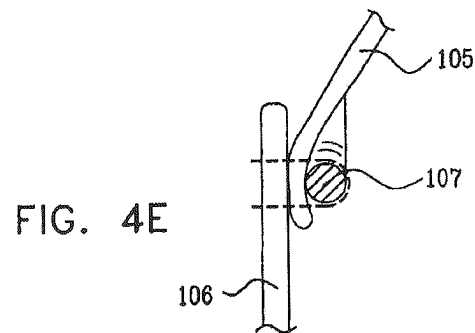
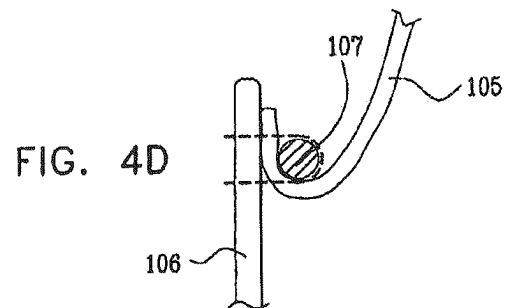
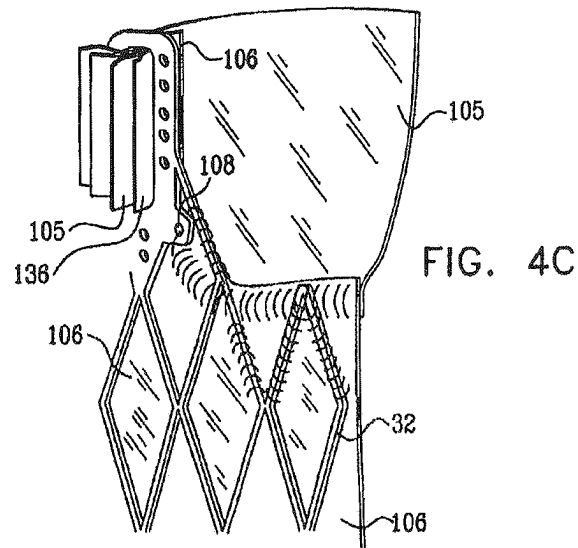


FIG. 5C

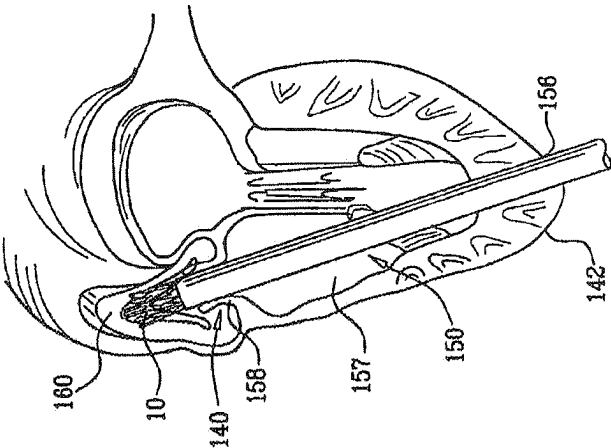


FIG. 5B

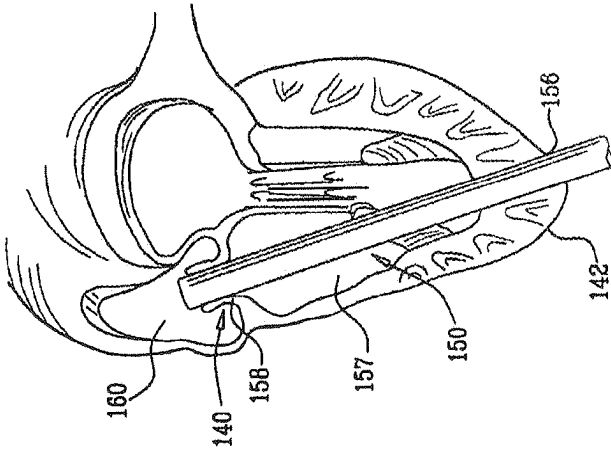
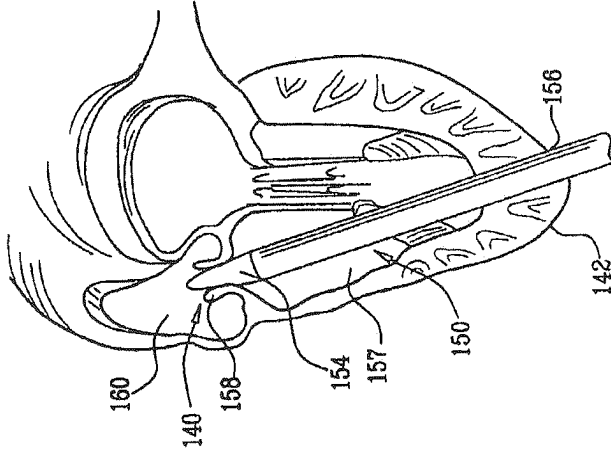
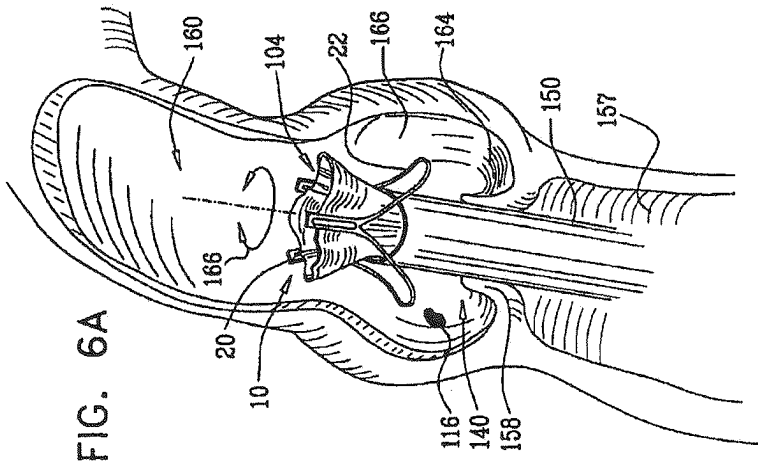
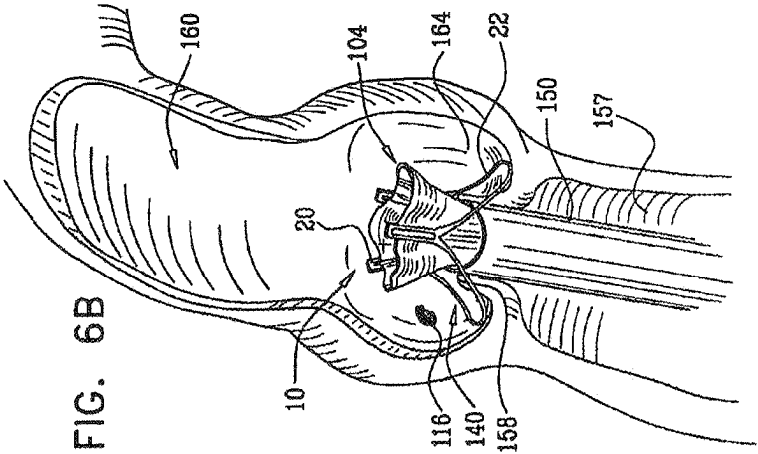


FIG. 5A





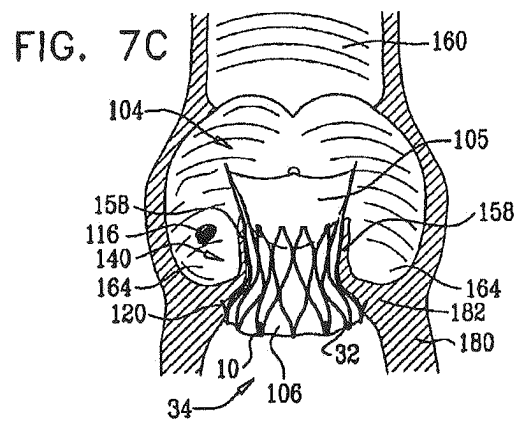
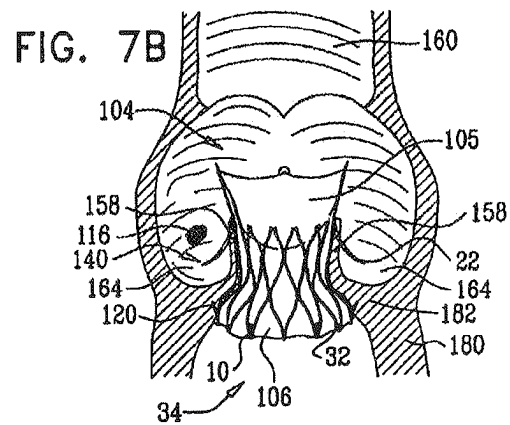
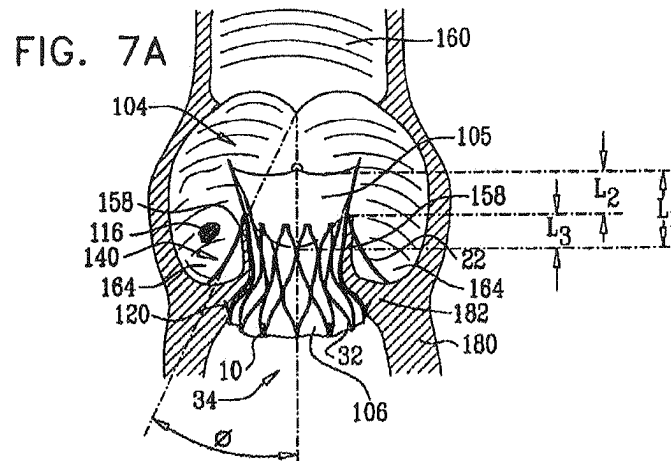


FIG. 7D

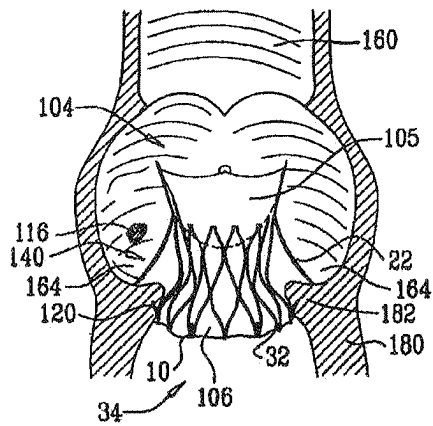


FIG. 7E

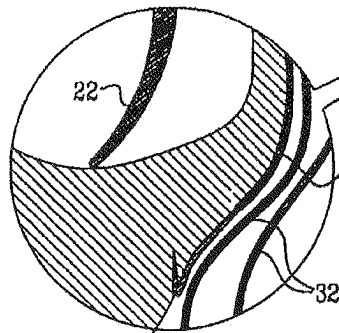
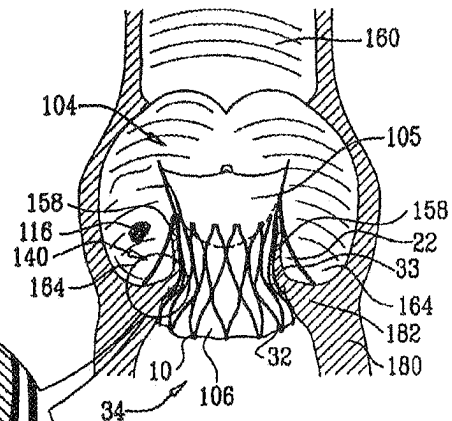


FIG. 8A

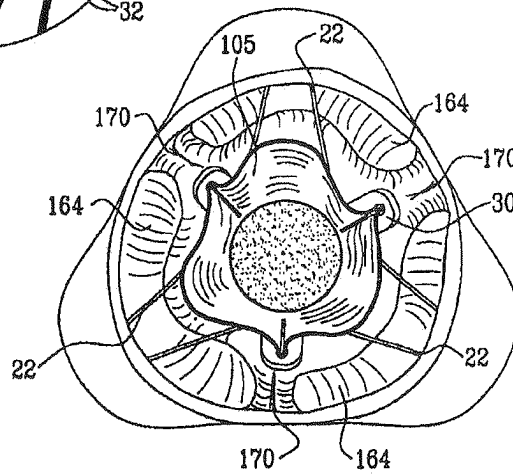


FIG. 8C

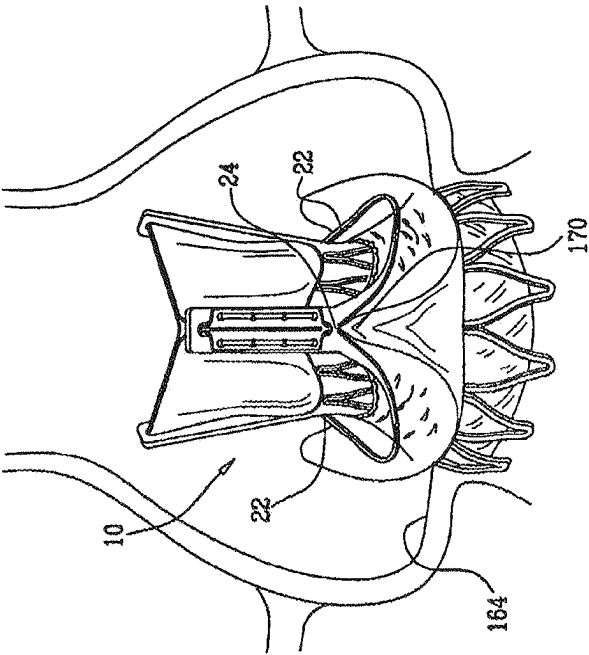


FIG. 8B

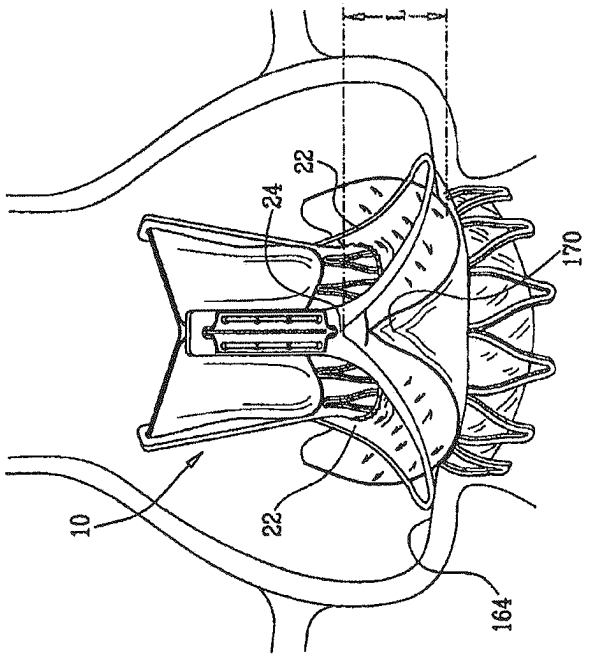


FIG. 9A

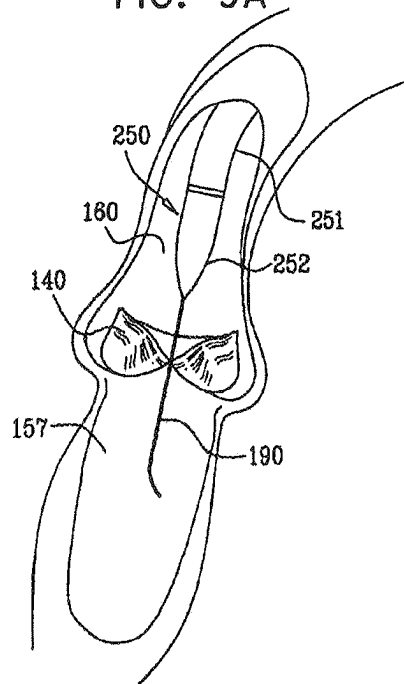


FIG. 9B

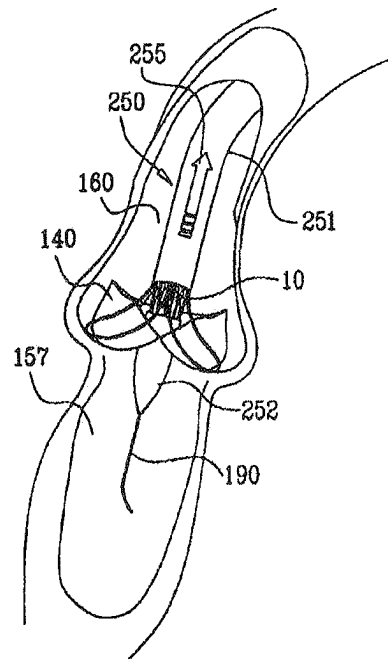


FIG. 9C

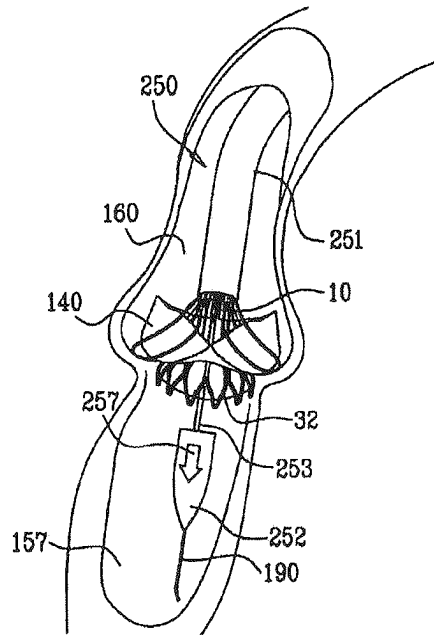
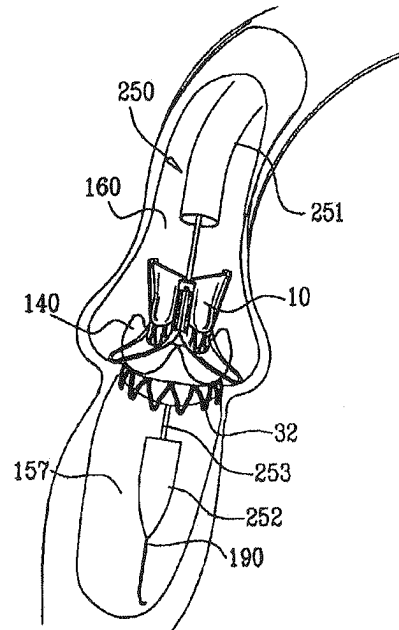


FIG. 9D



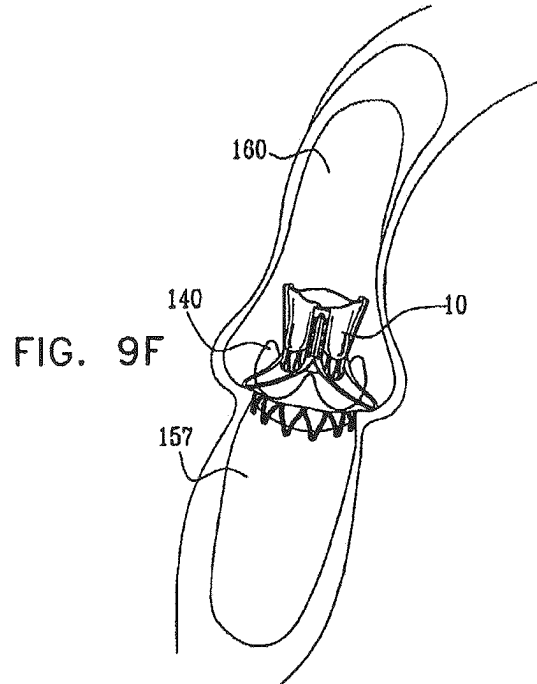
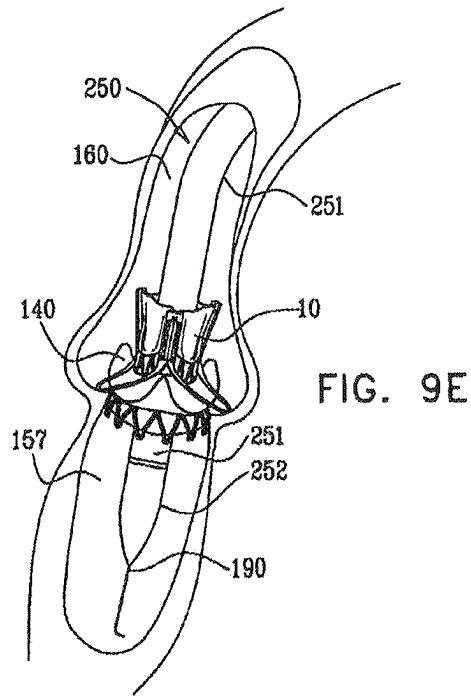


FIG. 10B

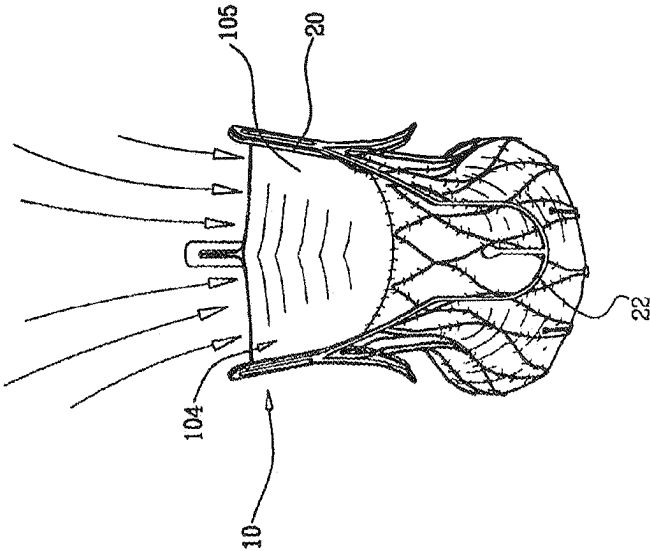


FIG. 10A

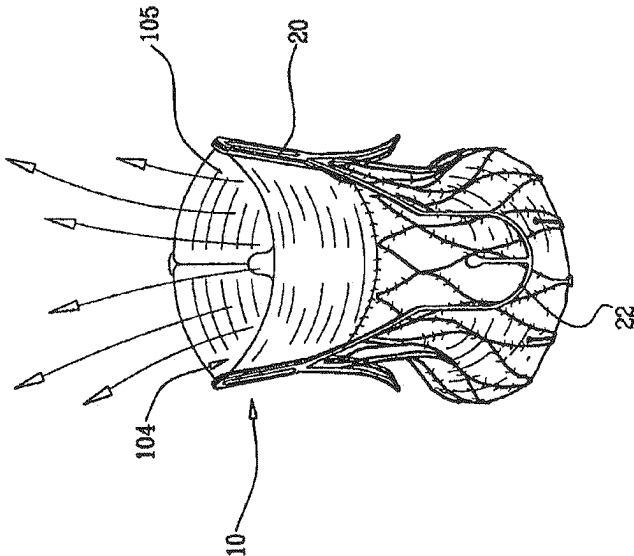


FIG. 11A

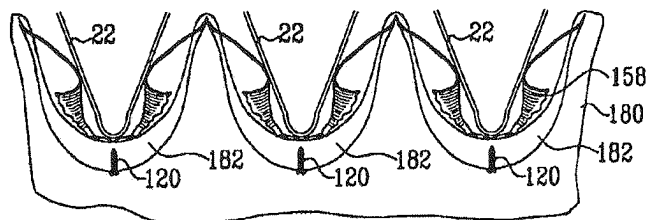


FIG. 11B

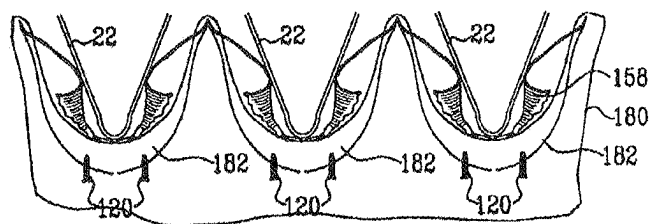


FIG. 11C

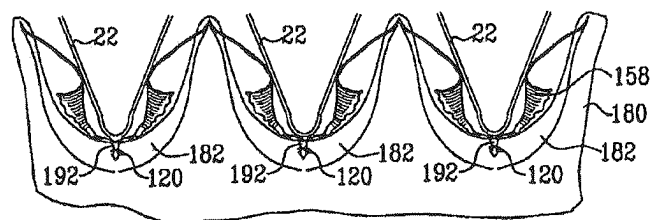
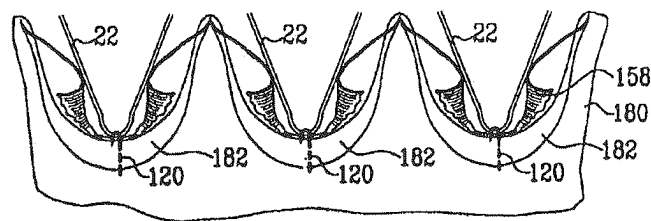


FIG. 11D



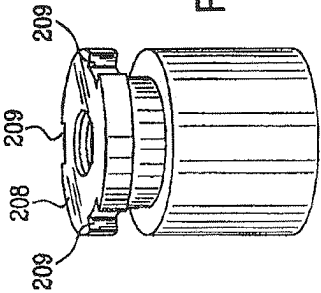
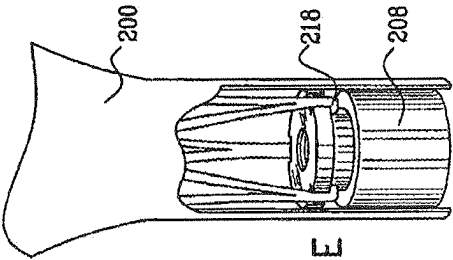
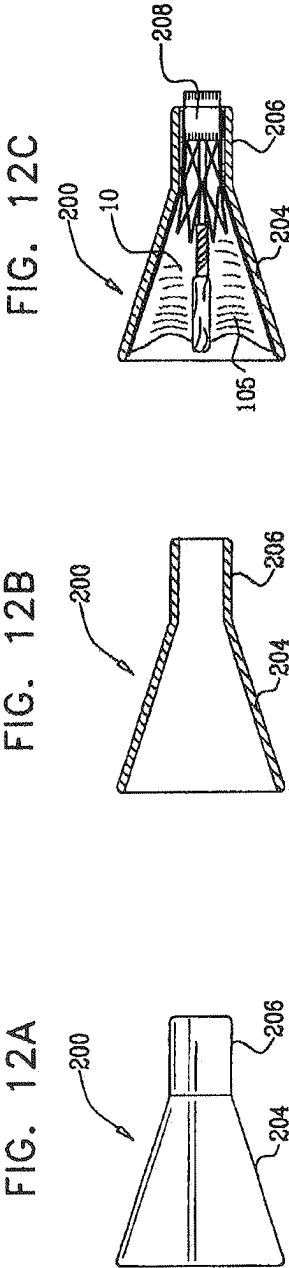


FIG. 12D

FIG. 12E

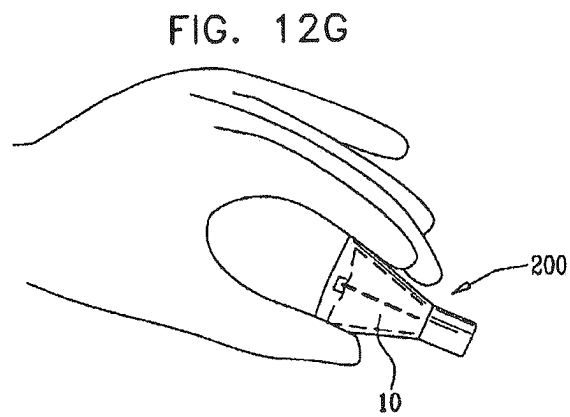
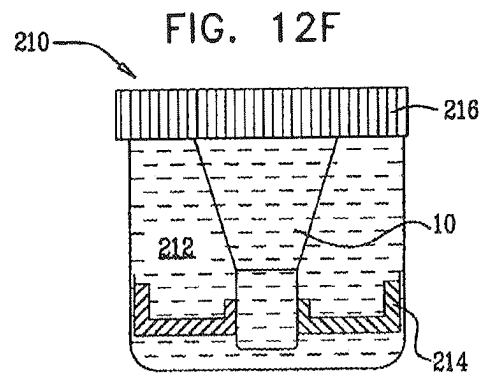


FIG. 13A

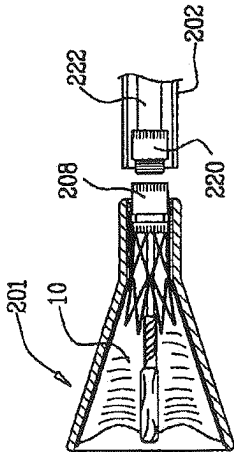


FIG. 13B

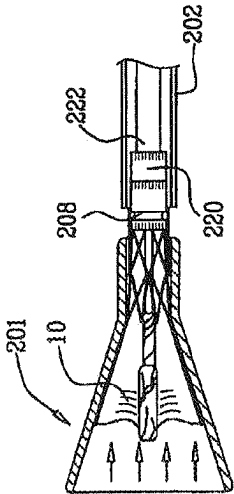


FIG. 13C

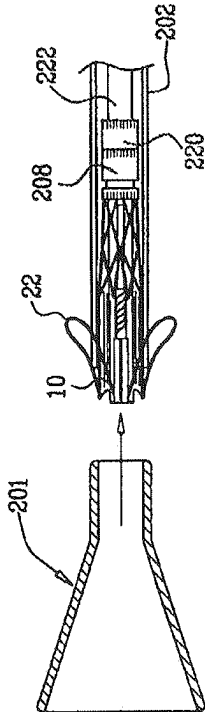


FIG. 13D

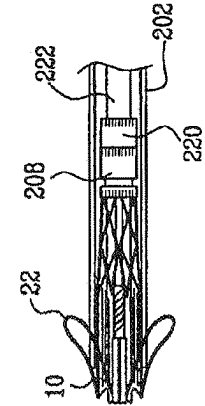


FIG. 14

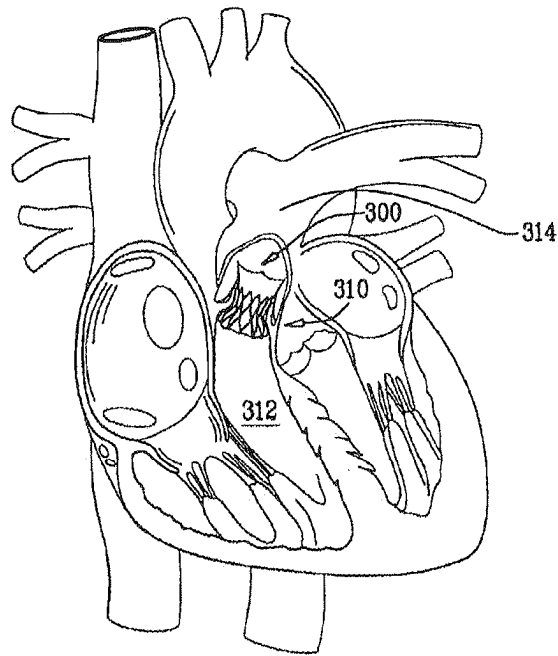
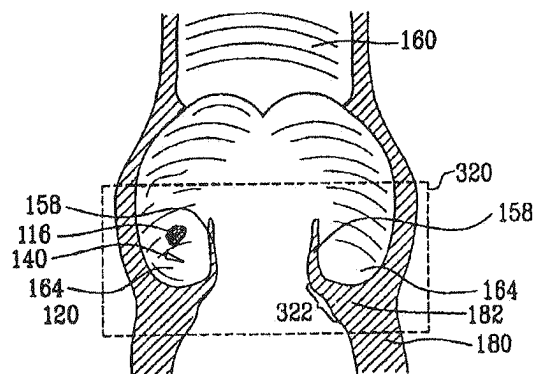


FIG. 15



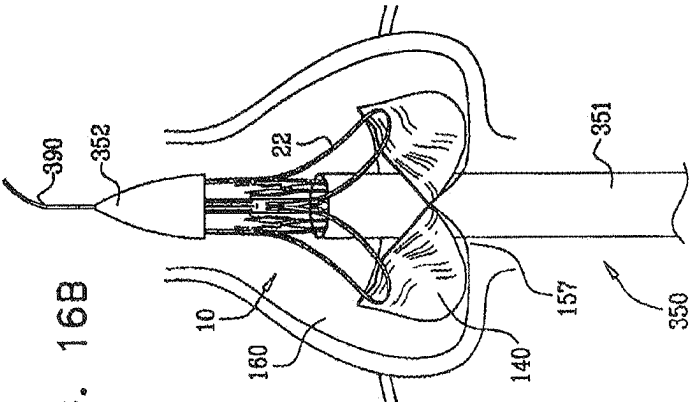


FIG. 16B

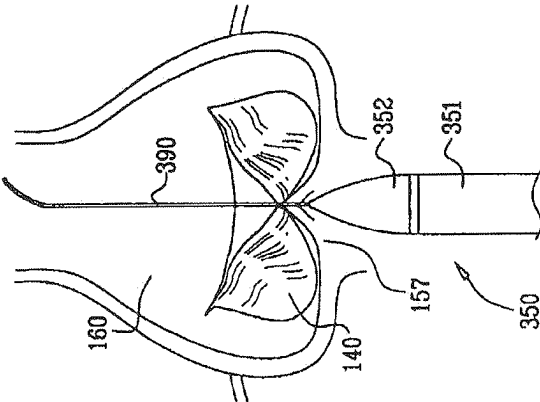


FIG. 16A

FIG. 16D

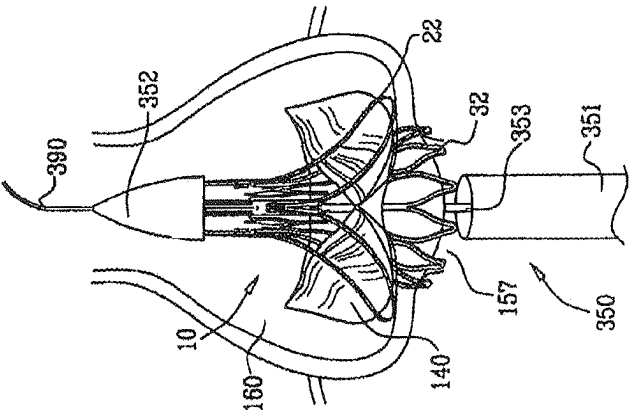


FIG. 16C

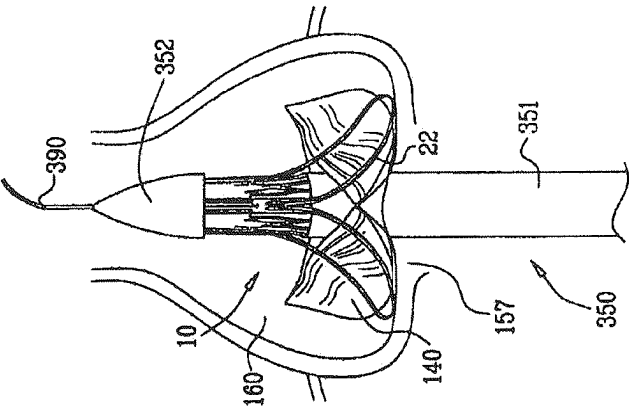


FIG. 16G

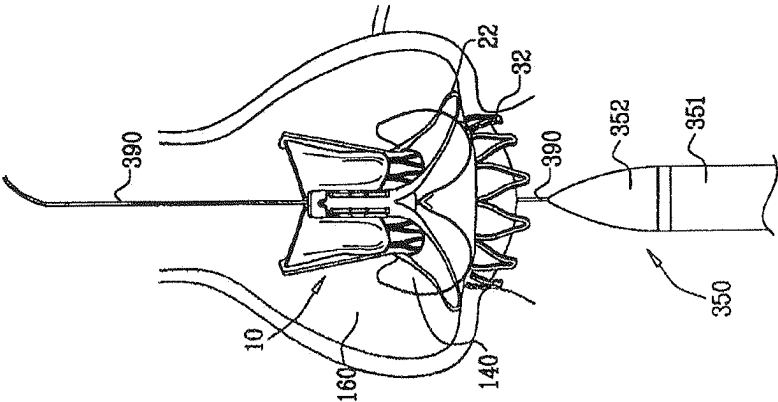


FIG. 16F

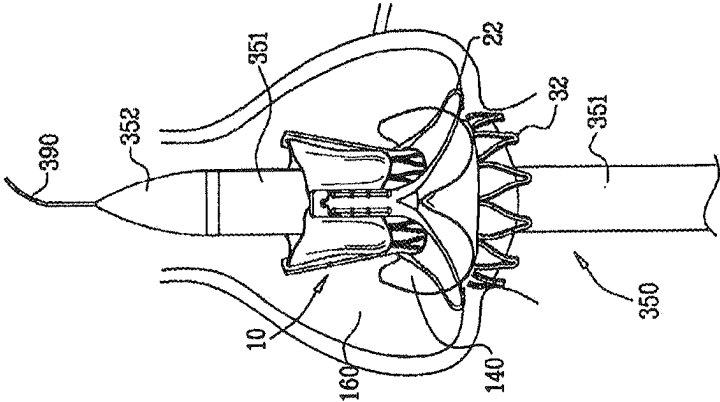
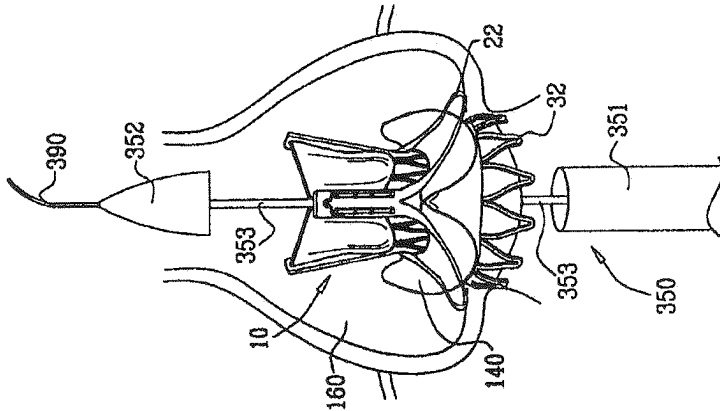
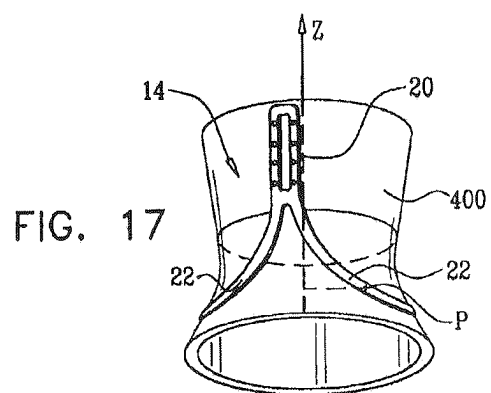
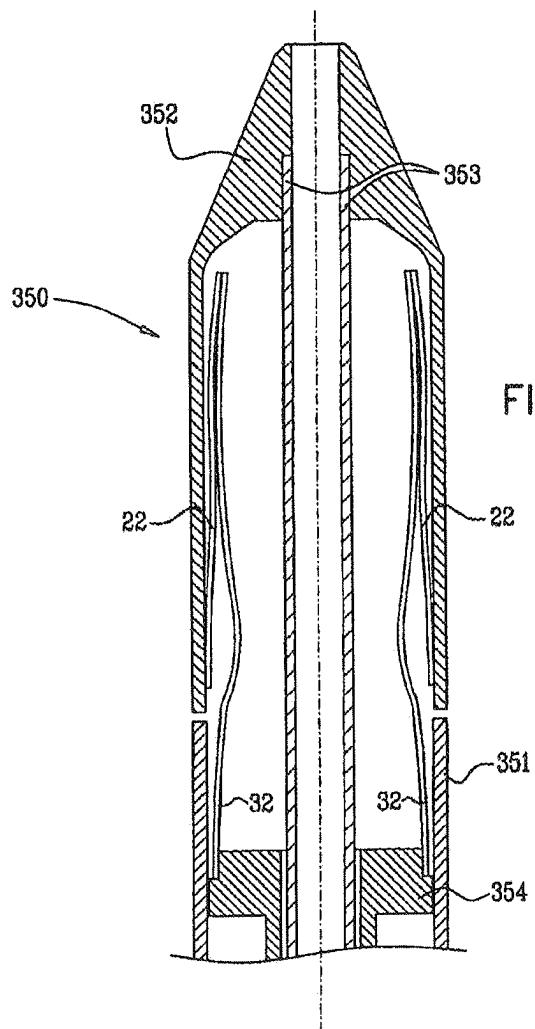


FIG. 16E





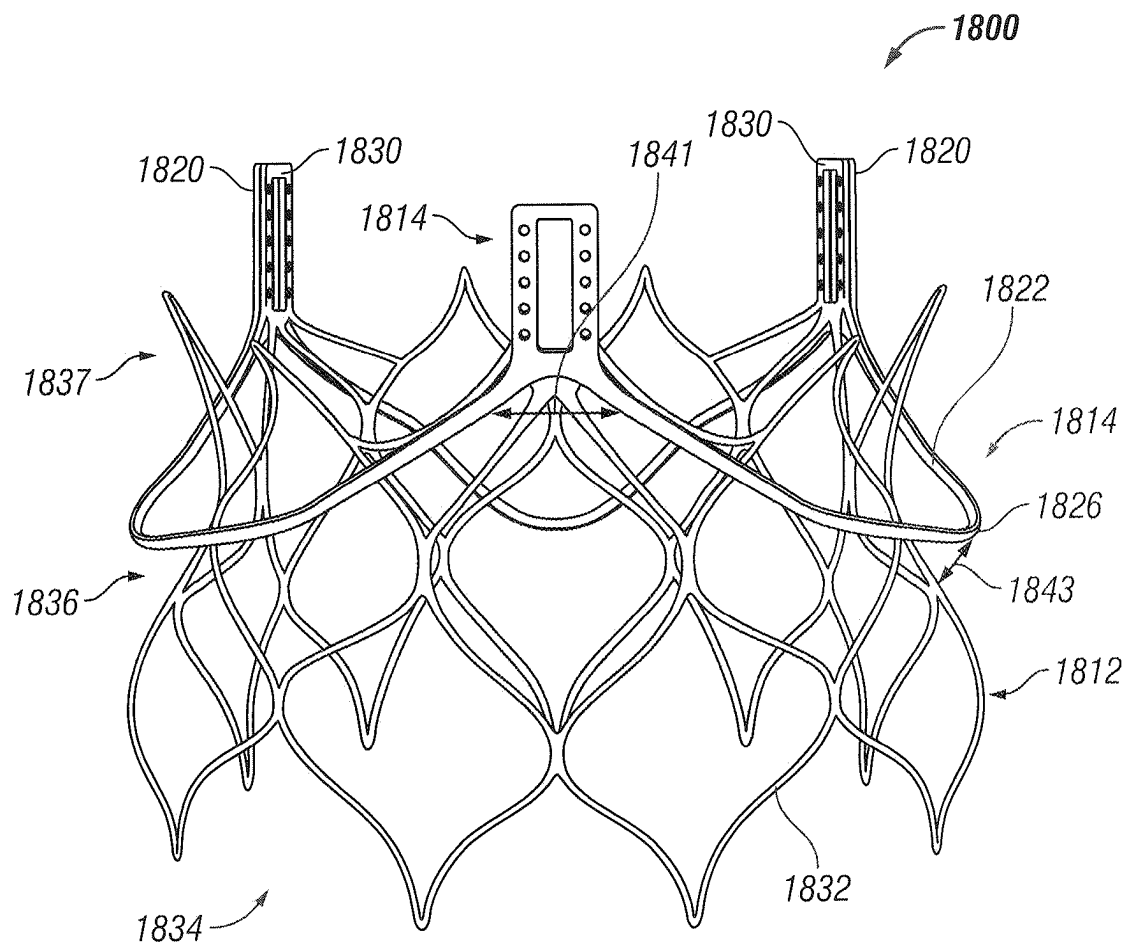


FIG. 18

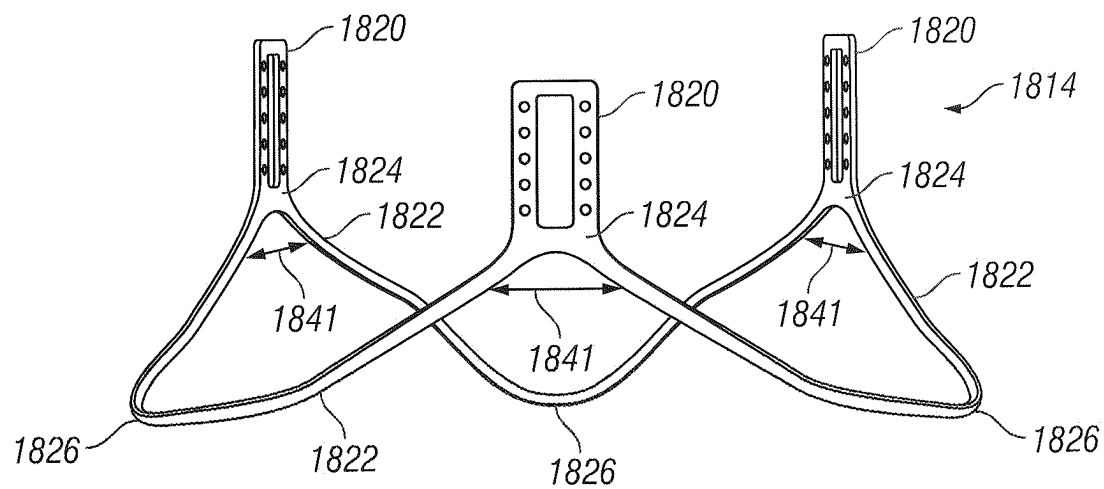


FIG. 19A

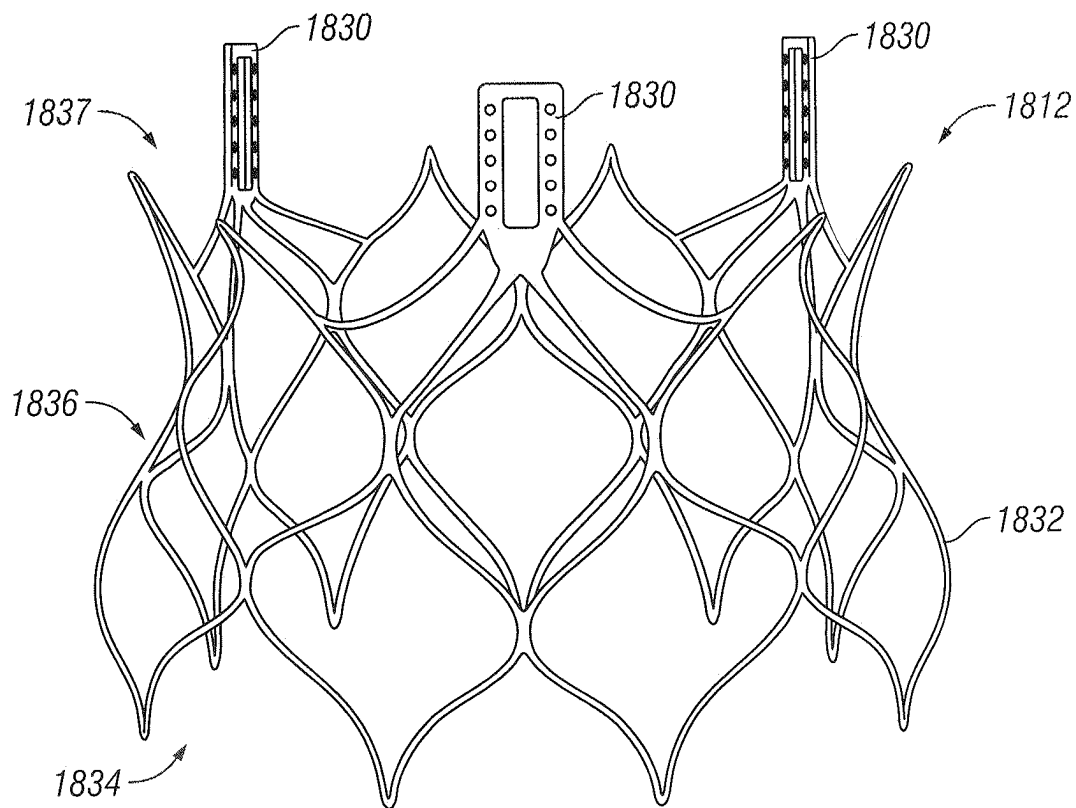


FIG. 19B

SINUS-ENGAGING VALVE FIXATION MEMBER

CROSS REFERENCE TO RELATED APPLICATIONS

The present application is a continuation of U.S. patent application Ser. No. 18/802,005, filed Aug. 13, 2025, now U.S. Pat. No. 12,257,148, which is a continuation of U.S. patent application Ser. No. 17/722,324, filed Apr. 16, 2022, now U.S. Pat. No. 12,076,237, which is a continuation of U.S. patent application Ser. No. 17/359,112, filed Jun. 25, 2021, now U.S. Pat. No. 11,304,802, which is a continuation of U.S. patent application Ser. No. 17/222,926, filed Apr. 5, 2021, now U.S. Pat. No. 11,304,801, which is continuation of U.S. patent application Ser. No. 16/724,358, filed Dec. 22, 2019, now U.S. Pat. No. 11,304,800, which is a continuation of U.S. patent application Ser. No. 15/881,853, filed Jan. 29, 2018, now U.S. Pat. No. 10,543,077, which is a continuation of U.S. patent application Ser. No. 15/179,483, filed Jun. 10, 2016, now U.S. Pat. No. 9,913,714, which is a continuation of U.S. patent application Ser. No. 14/484,387, filed Sep. 12, 2014, now U.S. Pat. No. 9,387,071, which is a continuation of U.S. patent application Ser. No. 12/721,789, filed Mar. 11, 2010, now U.S. Pat. No. 8,834,564, which is a continuation-in-part of U.S. patent application Ser. No. 11/726,893, filed Mar. 23, 2007, now U.S. Pat. No. 8,414,643, which claims the benefit of U.S. Provisional Application No. 60/845,728, filed Sep. 19, 2006, all of which are incorporated herein by reference in their entireties.

FIELD OF THE INVENTION

The present invention relates generally to prosthetic devices for the treatment of body lumens, and specifically to a valve prosthesis for such body lumens.

BACKGROUND OF THE INVENTION

PCT Publication WO 05/002466 to Schwammenthal et al., which is assigned to the assignee of the present application and is incorporated herein by reference, describes prosthetic devices for treating aortic stenosis.

PCT Publication WO 06/070372 to Schwammenthal et al., which is assigned to the assignee of the present application and is incorporated herein by reference, describes a prosthetic device having a single flow field therethrough, adapted for implantation in a subject, and shaped so as to define a fluid inlet and a diverging section, distal to the fluid inlet.

US Patent Application Publication 2006/0149360 to Schwammenthal et al., which is assigned to the assignee of the present application and is incorporated herein by reference, describes a prosthetic device including a valve-orifice attachment member attachable to a valve in a blood vessel and including a fluid inlet, and a diverging member that extends from the fluid inlet, the diverging member including a proximal end near the fluid inlet and a distal end distanced from the proximal end. A distal portion of the diverging member has a larger cross-sectional area for fluid flow therethrough than a proximal portion thereof.

U.S. Pat. No. 6,730,118 to Spencer et al., which is incorporated herein by reference, describes a valve prosthesis device suitable for implantation in body ducts. The device comprises a support stent, which comprises a deployable construction adapted to be initially crimped in a narrow configuration suitable for catheterization through the body

duct to a target location, and adapted to be deployed by exerting substantially radial forces from within by means of a deployment device to a deployed state in the target location; and a valve assembly comprising a flexible conduit having an inlet end and an outlet, made of pliant material attached to the support beams providing collapsible slack portions of the conduit at the outlet. The support stent is provided with a plurality of longitudinally rigid support beams of fixed length. When flow is allowed to pass through the valve prosthesis device from the inlet to the outlet, the valve assembly is kept in an open position, whereas a reverse flow is prevented as the collapsible slack portions of the valve assembly collapse inwardly providing blockage to the reverse flow.

U.S. Pat. No. 7,018,406 to Seguin et al., which is incorporated herein by reference, describes a prosthetic valve assembly for use in replacing a deficient native valve, comprising a replacement valve supported on an expandable valve support. If desired, one or more anchors may be used. The valve support, which entirely supports the valve annulus, valve leaflets, and valve commissure points, is configured to be collapsible for transluminal delivery and expandable to contact the anatomical annulus of the native valve when the assembly is properly positioned. The anchor engages the lumen wall when expanded and prevents substantial migration of the valve assembly when positioned in place. The prosthetic valve assembly is compressible about a catheter, and restrained from expanding by an outer sheath. The catheter may be inserted inside a lumen within the body, such as the femoral artery, and delivered to a desired location, such as the heart. When the outer sheath is retracted, the prosthetic valve assembly expands to an expanded position such that the valve and valve support expand within the deficient native valve, and the anchor engages the lumen wall.

U.S. Pat. No. 7,018,408 to Bailey et al., which is incorporated herein by reference, describes prosthetic cardiac and venous valves and a single catheter device, and minimally invasive techniques for percutaneous and transluminal valvuloplasty and prosthetic valve implantation. The device consists generally of a stent body member, a graft, and valve flaps. The graft is preferably a biocompatible, fatigue-resistant membrane which is capable of endothelialization, and is attached to the stent body member on at least portions of either or both the luminal and abluminal surfaces of the stent body member by suturing to or encapsulating stent struts. The valve leaflets are preferably formed by sections of the graft material attached to the stent body member. The stent body member is shaped to include the following stent sections: proximal and distal anchors, an intermediate annular stent section, and at least one valve arm or blood flow regulator struts.

U.S. Pat. No. 6,458,153 and US Patent Application Publication 2003/0023300 to Bailey et al., which are incorporated herein by reference, describe prosthetic cardiac and venous valves and a single catheter device, and minimally invasive techniques for percutaneous and transluminal valvuloplasty and prosthetic valve implantation.

US Patent Application Publication 2004/0186563 to Lobbi, which is incorporated herein by reference, describes a prosthetic heart valve having an internal support frame with a continuous, undulating leaflet frame defined therein. The leaflet frame has three cusp regions positioned at an inflow end intermediate three commissure regions positioned at an outflow end thereof. The leaflet frame may be cloth covered and flexible leaflets attached thereto form occluding surfaces of the valve. The support frame further

includes three cusp positioners rigidly fixed with respect to the leaflet frame and located at the outflow end of the support frame intermediate each pair of adjacent commissure regions. The valve is desirably compressible so as to be delivered in a minimally invasive manner through a catheter to the site of implantation. Upon expulsion from catheter, the valve expands into contact with the surrounding native valve annulus and is anchored in place without the use of sutures. In the aortic valve position, the cusp positioners angle outward into contact with the sinus cavities, and compress the native leaflets if they are not excised, or the aortic wall if they are. The support frame may be formed from a flat sheet of nitinol that is bent into a three-dimensional configuration and heat set. A holder having spring-like arms connected to inflow projections of the valve may be used to deliver, reposition and re-collapse the valve, if necessary.

US Patent Application Publication 2003/0130729 to Paniagua et al., which is incorporated herein by reference, describes a percutaneously implantable replacement heart valve device and a method of making same. The replacement heart valve device comprises a stent member made of stainless steel or self-expanding nitinol, and a biological tissue artificial valve means disposed within the inner space of the stent member. An implantation and delivery system has a central part which consists of a flexible hollow tube catheter that allows a metallic wire guide to be advanced inside it. The endovascular stented-valve is a glutaraldehyde fixed bovine pericardium which has two or three cusps that open distally to permit unidirectional blood flow.

US Patent Application Publication 2004/0236411 to Sarac et al., which is incorporated herein by reference, describes a prosthetic valve for replacing a cardiac valve, including an expandable support member and at least two valve leaflets made of a first layer of biological material selected from peritoneal tissue, pleural tissue, or pericardial tissue. A second layer of biological material is attached to the support member. The second layer is also made from peritoneal tissue, pleural tissue, or pericardial tissue. The second layer includes a radially inwardly facing surface that defines a conduit for directing blood flow. The valve leaflets extend across the conduit to permit unidirectional flow of blood through the conduit.

US Patent Application Publication 2005/0075720 to Nguyen et al., which is incorporated herein by reference, describes a method and system for minimally invasive replacement of a valve. The system includes a collapsible valve and anchoring structure, devices and methods for expanding the valve anchoring structure, adhesive means to seal the valve to the surrounding tissue, a catheter-based valve sizing and delivery system, native valve removal means, and a temporary valve and filter assembly to facilitate removal of debris material. The valve assembly comprises a valve and anchoring structure for the valve, dimensioned to fit substantially within the valve sinus.

US Patent Application Publication 2006/0058872 to Salahieh et al., which is incorporated herein by reference, describes an apparatus for endovascularly replacing a patient's heart valve. In some embodiments, the apparatus includes an expandable anchor supporting a replacement valve, the anchor and replacement valve being adapted for percutaneous delivery and deployment to replace the patient's heart valve, the anchor having a braid having atraumatic grasping elements adapted to grasp tissue in a vicinity of the patient's heart valve.

US Patent Application Publication 2005/0137688 to Salahieh et al., which is incorporated herein by reference, describes a method for percutaneously replacing a heart

valve of a patient. In some embodiments the method includes the steps of percutaneously delivering a replacement valve and an expandable anchor to a vicinity of the heart valve in an unexpanded configuration; expanding the anchor to a deployed configuration in which the anchor contacts tissue at a first anchor site; repositioning the anchor to a second anchor site; and deploying the anchor at the second anchor site.

US Patent Application Publication 2005/0137690 to Salahieh et al., which is incorporated herein by reference, describes apparatus for endovascularly replacing a patient's heart valve, including: a delivery catheter having a diameter of 21 french or less; an expandable anchor disposed within the delivery catheter; and a replacement valve disposed within the delivery catheter. The invention also includes a method for endovascularly replacing a heart valve of a patient. In some embodiments the method includes the steps of: inserting a catheter having a diameter no more than 21 french into the patient; endovascularly delivering a replacement valve and an expandable anchor to a vicinity of the heart valve through the catheter; and deploying the anchor and the replacement valve.

US Patent Application Publication 2005/0137691 to Salahieh et al., which is incorporated herein by reference, describes apparatus for endovascularly replacing a patient's heart valve, including: a custom-designed anchor; and a replacement valve, wherein the custom-designed anchor is adapted to engage native leaflets of the heart valve, and wherein the anchor and the valve are adapted for in vivo expansion and coupling to one another to form composite apparatus that endovascularly replaces the heart valve. The invention also includes a method for endovascularly replacing a patient's heart valve. In some embodiments the method includes the steps of: providing apparatus comprising an anchor piece and a replacement valve piece; endovascularly delivering the anchor piece to a vicinity of the heart valve in a collapsed delivery configuration; expanding the anchor piece to a deployed configuration; engaging at least one valve leaflet of the heart valve with the anchor piece; endovascularly delivering the replacement valve piece to the vicinity of the heart valve in a collapsed delivery configuration; expanding the replacement valve piece to a deployed configuration; and coupling the valve piece to the anchor piece in vivo to form composite two-piece apparatus that endovascularly replaces the patient's heart valve.

US Patent Application Publication 2005/0137695 to Salahieh et al., which is incorporated herein by reference, describes apparatus for endovascularly replacing a patient's heart valve, including a replacement valve adapted to be delivered endovascularly to a vicinity of the heart valve; an expandable anchor adapted to be delivered endovascularly to the vicinity of the heart valve; and a lock mechanism configured to maintain a minimum amount of anchor expansion.

US Patent Application Publication 2005/0143809 to Salahieh et al., which is incorporated herein by reference, describes techniques for endovascularly replacing a heart valve of a patient. One aspect described is a method including the steps of endovascularly delivering a replacement valve and an expandable anchor to a vicinity of the heart valve in an unexpanded configuration; and applying an external non-hydraulically expanding or non-pneumatically expanding actuation force on the anchor to change the shape of the anchor, such as by applying proximally and/or distally directed force on the anchor using a releasable deployment tool to expand and contract the anchor or parts of the anchor. Another aspect described includes an apparatus including a

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replacement valve; an anchor; and a deployment tool comprising a plurality of anchor actuation elements adapted to apply a non-hydraulically expanding or non-pneumatically expanding actuation force on the anchor to reshape the anchor.

US Patent Application Publication 2005/0182483 to Osborne et al., which is incorporated herein by reference, describes a venous valve prosthesis having a substantially non-expandable, valve portion comprising a valve-closing mechanism, such as a pair of opposing leaflets; and an anchoring portion, such as one or more self-expanding frames or stents that are expandable to anchor the prosthesis at the implantation site. In one embodiment, the rigid valve portion includes a deposition of material such as pyrolytic carbon to reduce the thrombogenicity of the blood-contacting surfaces. The anchoring portions preferably include a covering, such as a tubular construct of synthetic or collagen-derived material (such as a bioremodelable ECM material), which attaches about the support structure such that blood flow is directed through the valve mechanism as it transitions from the larger diameter anchoring portion to the intermediate, smaller-diameter portion of the prosthesis. In another embodiment, the valve support housing and valve-closing elements are delivered in a collapsed, folded, and/or disassembled state sized for delivery, then manipulated in situ to the second expanded configuration following deployment.

US Patent Application Publication 2005/0197695 to Stacchino et al., which is incorporated herein by reference, describes a cardiac-valve prosthesis adapted for percutaneous implantation. The prosthesis includes an armature adapted for deployment in a radially expanded implantation position, the armature including a support portion and an anchor portion, which are substantially axially coextensive with respect to one another. A set of leaflets is coupled to the support portion. The leaflets can be deployed with the armature in the implantation position. The leaflets define, in the implantation position, a flow duct that is selectively obstructable. The anchor portion can be deployed to enable anchorage of the cardiac-valve prosthesis at an implantation site.

US Patent Application Publication 2005/0240200 to Berghem, which is incorporated herein by reference, describes methods and systems for introducing a delivery device in the heart at or near the apex of the heart, wherein the methods include advancing the prosthesis to a target site, and disengaging the prosthesis from the delivery device at the target site for implantation. Specifically, the valve replacement systems are described for delivering a replacement heart valve to a target site in or near a heart. The valve replacement system comprises a trocar or other suitable device to penetrate the heart at or near the apex of the heart, a delivery member that is movably disposed within the trocar, and a replacement cardiac valve disposed on the delivery member. The delivery member may further comprise mechanical or inflatable expanding members to facilitate implantation of the prosthetic valve at the target site.

US Patent Application Publication 2006/0025857 to Berghem et al., which is incorporated herein by reference, describes valve prostheses adapted to be initially crimped in a narrow configuration suitable for catheterization through body ducts to a target location, and adapted to be deployed by exerting substantially radial forces from within by means of a deployment device to a deployed state in the target location.

US Patent Application Publication 2006/0025855 to Lashinski et al., which is incorporated herein by reference, describes a cardiovascular prosthetic valve comprising an

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inflatable body that has at least a first inflatable chamber and a second inflatable chamber that is not in fluid communication with the first inflatable chamber. The inflatable body is configured to form, at least in part, a generally annular ring.

5 A valve is coupled to the inflatable body. The valve is configured to permit flow in a first axial direction and to inhibit flow in a second axial direction opposite to the first axial direction. A first inflation port is in communication with the first inflatable chamber. A second inflation port in communication with the second inflatable chamber.

US Patent Application Publication 2006/0047338 to Jensen et al., which is incorporated herein by reference, describes a cardiac valve having a support frame having a first end member and a second end member opposing the first end member in a substantially fixed distance relationship, and a cover extending over the support frame to allow for unidirectional flow of a liquid through the valve.

US Patent Application Publication 2006/0052867 to Revuelta et al., which is incorporated herein by reference, describes a method for functionally replacing a previously implanted prosthetic heart valve. The method includes positioning a replacement prosthetic heart valve within an internal region defined by the previously implanted prosthetic heart valve. The replacement prosthetic heart valve is then physically docked to the previously implanted prosthetic heart valve. With this technique, the previously implanted prosthetic heart valve serves as a platform for securement of the replacement prosthetic heart valve to the patient's native tissue.

US Patent Application Publication 2006/0074485 to Realyvasquez, which is incorporated herein by reference, describes methods and apparatus for valve repair or replacement. In one embodiment, the apparatus is a valve delivery device comprising a first apparatus and a second apparatus. The first apparatus includes a heart valve support having a proximal portion and a distal portion and a heart valve excisor slidably mounted on said first apparatus. The second apparatus includes a fastener assembly having a plurality of penetrating members mounted to extend outward when the assembly assumes an expanded configuration; and a heart valve prosthesis being releasably coupled to said second apparatus. The first apparatus and second apparatus are sized and configured for delivery to the heart through an opening formed in a femoral blood vessel. The heart valve prosthesis support is movable along a longitudinal axis of the device to engage tissue disposed between the anvil and the valve prosthesis.

US Patent Application Publication 2006/0259136 to Nguyen et al., which is incorporated herein by reference, describes a heart valve prosthesis having a self-expanding multi-level frame that supports a valve body comprising a skirt and plurality of coapting leaflets. The frame transitions between a contracted delivery configuration that enables percutaneous transluminal delivery, and an expanded deployed configuration having an asymmetric hourglass shape. The valve body skirt and leaflets are constructed so that the center of coaptation may be selected to reduce horizontal forces applied to the commissures of the valve, and to efficiently distribute and transmit forces along the leaflets and to the frame. Alternatively, the valve body may be used as a surgically implantable replacement valve prosthesis.

U.S. Pat. No. 7,137,184 to Schreck, which is incorporated herein by reference, describes methods for forming a support frame for flexible leaflet heart valves from a starting blank include converting a two-dimensional starting blank into the three-dimensional support frame. The material may be

superelastic, such as NITINOL, and the method may include bending the 2-D blank into the 3-D form and shape setting it. A merely elastic material such as ELGILOY may be used and plastically deformed in stages, possibly accompanied by annealing, to obtain the 3-D shape.

U.S. Pat. No. 6,558,418 to Carpentier et al., which is incorporated herein by reference, describes a highly flexible tissue-type heart valve is disclosed having a structural stent in a generally cylindrical configuration with cusps and commissures that are permitted to move radially. The stent commissures are constructed so that the cusps are pivotably or flexibly coupled together at the commissures to permit relative movement therebetween. The stent may be cloth-covered and may be a single element or may be made in three separate elements for a three cusp valve, each element having a cusp portion and two commissure portions; adjacent commissure portions for each pair of adjacent stent element combining to form the stent commissures. If the stent has separate elements their commissure portions may be pivotably or flexibly coupled, or may be designed to completely separate into independent leaflets at bioresorbable couples. The cloth covering may have an outwardly projecting flap that mates with valve leaflets (e.g., pericardial leaflets) along the cusps and commissures. A connecting band may be provided that follows the cusps and commissures and extends outwardly. The valve is connected to the natural tissue along the undulating connecting band using conventional techniques, such as sutures.

U.S. Pat. No. 6,296,662 to Caffey, which is incorporated herein by reference, describes heart valve prosthesis including a heart valve formed of a flexible material. An elongated stent member is provided in the valve and includes terminal ends. A plurality of flexible post members are formed in the stent member. Each post member includes a pair of opposite sides. A crimp collar interconnects the terminal ends of the stent member. The crimp collar is positioned between adjacent post members. A first radius is formed in the stent member between the crimp collar and an adjacent side of each adjacent post member. A plurality of second radii are formed in the stent member between an opposite side of a first one of the adjacent post members and an opposite side of a second one of the adjacent post members. The second radii are greater than each first radius.

The following patents and patent application publication, all of which are incorporated herein by reference, may be of interest:

U.S. Pat. No. 6,312,465 to Griffin et al.

U.S. Pat. No. 5,908,451 to Yeo

U.S. Pat. No. 5,344,442 to Deac

U.S. Pat. No. 5,354,330 to Hanson

US Patent Application Publication 2004/0260389 to Case et al.

SUMMARY OF THE INVENTION

In some embodiments of the present invention, an aortic valve prosthesis for treating a native stenosed valve comprises two portions that are configured to axially sandwich a native valve complex from the aortic (i.e., downstream) and left-ventricular (i.e., upstream) sides thereof, and a collapsible valve that is configured to be open during systole and closed during diastole. The two portions typically include a collapsible inner support structure that serves as a proximal (i.e., upstream) fixation member, and a collapsible outer support structure that serves as a distal (i.e., downstream) fixation member. The distal fixation member is configured to be positioned in an ascending aorta of the

subject, and to apply, to an aortic side of the native valve complex, a first axial force directed toward a left ventricle of the subject. The proximal fixation member is configured to be positioned at least partially on the left-ventricular side of the aortic valve, typically extending at least partially into the left ventricular outflow tract (LVOT), and to apply, to a left-ventricular side of the aortic annulus (typically, at the top of the left ventricle), a second axial force directed in a downstream direction (i.e., toward the ascending aorta). Application of the first and second forces couples the prosthesis to the native valve.

In some embodiments of the present invention, the valve prosthesis is configured to treat a native pulmonary valve.

For some applications, the distal fixation member is shaped so as to define engagement arms that are configured to be positioned distal to the native annulus, at least partially within the aortic sinuses, and, for some applications, to apply the first axial force. Typically, for these applications, the distal fixation member is configured to apply the first axial force to the floors of the aortic sinuses.

The valve prosthesis is configured to be placed in the native stenosed valve using a minimally-invasive approach, such as an endovascular or transapical approach. The valve prosthesis is configured to be self-expanding and easy to position, and typically does not require suturing to be held in place. The native valve leaflets typically do not need to be opened to the maximal extent possible, but rather only to the extent which allows insertion of the narrowest part of the valve prosthesis, the diameter of which is typically about 15-20 mm. Placement of the valve prosthesis is thus accompanied by reduced risk of embolism of calcific or thrombotic material dislodged from the valve and coronary occlusion compared to many conventional valve prosthesis implantation procedures.

Unlike some valve prostheses known in the art, the valve prosthesis of some embodiments of the present invention does not rely for fixation on high forces applied outwardly radially against the native valve. Typically, a ratio of (a) the first or second axial force applied by the valve prosthesis to (b) the radial force applied outwardly by the valve prosthesis against the native valve is greater than 1.5:1, e.g., greater than 3:1 or greater than 6:1. For some applications, the valve prosthesis applies a radial force of less than 0.5 pounds (0.23 kilogram-force) outwardly against the native valve, such as less than 0.3 pounds (0.14 kgf), or less than 0.1 pounds (0.045 kgf). For some applications, the valve prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole, and the second axial force with a force of at least 1 g (e.g., at least 5 g) during systole. For some applications, the valve prosthesis is configured to apply the first axial force with a force of no more than 1700 g during diastole.

In other embodiments, the valve prosthesis applies a force outwardly radially against the native valve that is sufficient to aid with fixation of the prosthesis, or sufficient to fixate the prosthesis.

In some embodiments of the present invention, the valve prosthesis applies such outwardly radial forces only to the extent necessary to allow insertion of the prosthesis through the native valve, but not sufficiently to fully open the native leaflets to the maximum extent possible. This level of radial force application, typically in conjunction with the distal fixation member placed upon the aortic side of the native valve leaflets, prevents pushing of the native valve leaflets against the coronary ostia. Additionally, the configuration of the valve prosthesis generally reduces or eliminates leakage around the prosthetic valve, by avoiding damage to the

native leaflets. Such damage is avoided because the valve prosthesis typically does not fully open, fold over, or crimp the native leaflets. Instead, the valve prosthesis gently envelops the leaflets between the distal fixation member (e.g., the engagement arms thereof) and the proximal fixation member. Such damage to the native leaflets is also avoided because the valve prosthesis typically does not apply substantial axial force to the native valve commissures. Furthermore, for applications in which the valve prosthesis comprises a bulging proximal skirt, as described hereinbelow, the skirt generally helps reduce leakage around the prosthetic valve.

Typically, the valve prosthesis does not apply an axial force to the tips of the native valve leaflets that would result in shortening of the length of the leaflets, or forced bending, crimping, or folding over of the leaflets. Given the complex composition of the leaflets (fibrous tissue, soft atheroma, and calcifications), such compression might result in the application of shear forces to the leaflets, which might dislodge material and cause an embolism.

Although the valve prosthesis is generally described herein with respect to treating a native aortic valve, in some embodiments the valve prosthesis is used to treat a native pulmonary valve (i.e., the other semilunar valve in the heart), or another native valve of the body, with appropriate modifications to the valve prosthesis.

As used herein, including in the claims, the “native valve complex” includes the native semilunar valve leaflets, the annulus of the valve, the subvalvular tissue on the ventricular side, and the lower half of the semilunar sinuses.

There is therefore provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the native valve complex having three semilunar sinuses and three native commissures, the prosthesis including a valve prosthesis support, which includes a support structure including exactly three engagement arms that meet one another at three respective junctures,

wherein the engagement arms are shaped so as define three peak complexes at the three respective junctures, and three trough complexes, each of which is between two of the peak complexes, and

wherein upon implantation of the prosthesis, each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses, such that each of the peak complexes is disposed distal to and in rotational alignment with a respective one of the native commissures, and each of the trough complexes is disposed at least partially within the respective one of the semilunar sinuses.

In an embodiment, the native semilunar valve includes a native aortic valve of the subject, the semilunar sinuses include respective aortic sinuses, and upon implantation of the prosthesis, each of the engagement arms is disposed at least partially within the respective one of the aortic sinuses.

In an embodiment, the native semilunar valve includes a native pulmonary valve of the subject, the semilunar sinuses include respective pulmonary sinuses, and upon implantation of the prosthesis, each of the engagement arms is disposed at least partially within the respective one of the pulmonary sinuses.

In an embodiment, the engagement arms are shaped such that each of the peak complexes includes exactly one peak at its respective one of the junctures. In an embodiment, the engagement arms are shaped such that each of the trough complexes includes exactly one trough.

For some applications, the engagement arms are shaped so as to define exactly one trough between each two of the peak complexes. Alternatively, the engagement arms are shaped so as to define a plurality of troughs between each two of the peak complexes.

In an embodiment, the engagement arms are configured to touch respective transitions between the respective semilunar sinuses and respective native leaflet roots of the native valve complex, upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured such that, during implantation of the prosthesis, the peak complexes self-align with the respective native commissures.

For some applications, upon implantation of the prosthesis, each of the peak complexes is disposed in the rotational alignment with the respective one of the native commissures with a rotational offset. Alternatively, upon implantation of the prosthesis, each of the peak complexes is disposed in the rotational alignment with the respective one of the native commissures without a rotational offset.

In an embodiment, the valve prosthesis support, upon implantation of the prosthesis, does not press upon the native commissures of the native semilunar valve. Alternatively, the peak complexes, upon implantation of the prosthesis, touch the respective native commissures of the native semilunar valve at the respective junctures of the engagement arms.

For some applications, the prosthesis is configured to apply a radial force of less than 0.5 pounds outwardly against the native semilunar valve.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve.

For some applications, the prosthesis is configured, upon implantation thereof, such that the engagement arms apply a force to distal sides of the leaflets of the native semilunar valve while the engagement arms are generally parallel to the distal sides of the leaflets.

In an embodiment, the valve prosthesis support is configured such that, upon implantation of the prosthesis, the valve prosthesis support does not fold over leaflets of the native semilunar valve. In an embodiment, the valve prosthesis support is configured such that, upon implantation of the prosthesis, the valve prosthesis support does not push leaflets of the native semilunar valve towards respective semilunar sinus floors of the native valve complex. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex. In an embodiment, the valve prosthesis support is configured to elevate leaflets of the native semilunar valve from within the semilunar sinuses upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the engagement arms are aligned by rotation with respective ones of the semilunar sinuses.

In an embodiment, each of the engagement arms includes at least one extension element that extends from the engagement arm, which at least one extension element is configured to engage a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis.

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In an embodiment, each of the engagement arms is configured to engage a respective one of the semilunar sinuses upon implantation of the prosthesis. For some applications, each of the engagement arms is configured to firmly engage the respective one of the semilunar sinuses upon implantation of the prosthesis.

In an embodiment, the valve prosthesis support is configured not to apply a force to leaflets of the native semilunar valve sufficient to hold the prosthesis in place.

For some applications, each of the engagement arms is shaped so as to define at least one extension element that extends from the engagement arm, and each of the engagement arms and its respective at least one extension element are configured such that the engagement arm engages, via the at least one extension element, a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis.

For some applications, each of the engagement arms is shaped to define a length, parallel to a longitudinal axis of the prosthesis, between (a) at least one of the junctures and (b) a contact point of one of the engagement arms that meets at the juncture with a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis, which length is greater than 6 mm.

In an embodiment, the prosthesis includes a prosthetic valve including one or more prosthetic leaflets, at least a portion of each of the prosthetic leaflets is configured to assume a closed position during diastole and an open position during systole, and the at least a portion is not directly coupled to any of the engagement arms. For some applications, the prosthetic valve is coupled to the support structure such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve, upon implantation of the prosthesis. For some applications, the prosthetic valve includes a collapsible pliant material, configured to assume the open and closed positions. For some applications, the valve prosthesis support and the prosthetic valve are configured to define a single flow field through the valve prosthesis support and the prosthetic valve. Alternatively, the valve prosthesis support and the prosthetic valve are configured to define a plurality of flow fields through the valve prosthesis support and the prosthetic valve.

In an embodiment, the support structure includes exactly three commissural posts, to which the junctures of the engagement arms are respectively attached. For some applications, upon implantation of the prosthesis, the commissural posts are rotationally aligned with respective ones of the native commissures.

In an embodiment, the engagement arms are shaped so as to flare out laterally to an angle with respect to a central axis of the prosthesis. In an embodiment, the engagement arms conform to a shape of a semilunar root of the native valve complex when the engagement arms are flared out. In an embodiment, the engagement arms are shaped so as to curve outwards laterally. In an embodiment, a shape of at least one of the engagement arms is generally characterized by a function $z''(r) \geq 0$, where z is a height of any given point on the at least one engagement arm measured along a longitudinal axis of the prosthesis, and r is a distance from the longitudinal axis to the given point. For some applications, the shape is generally characterized by the function $z''(r) > 0$.

In an embodiment, the support structure is configured to serve as a distal fixation member, the valve prosthesis support includes a proximal fixation member, and the proximal fixation member and the engagement arms of the distal fixation member are configured to axially sandwich the

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native valve complex from ventricular and downstream sides thereof, respectively, upon implantation of the prosthesis.

In an embodiment, the engagement arms are configured to be disposed, during an implantation procedure, at least partially within the respective ones of the semilunar sinuses before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex, such that the arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount, the opening being because of force applied by the proximal fixation member to the leaflets.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the ventricular side of the native valve complex. For some applications, the at least one barb is configured to pierce the ventricular side of the native valve complex. Alternatively, the at least one barb is configured to protrude into tissue of the ventricular side of the native valve complex, without piercing the tissue. In an embodiment, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible. For some applications, the distal fixation member is configured to be positioned, during an implantation procedure, in a downstream artery while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex, the downstream artery selected from the group consisting of: an ascending aorta, and a pulmonary trunk. For some applications, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure. For some applications, the inner and outer support structures are configured to be coupled to one another during an implantation procedure.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which the engagement arms extend radially outward. In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the strut supports are aligned with the respective native commissures. In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply an axial force directed toward a downstream artery selected from the group consisting of: an ascending aorta, and a pulmonary trunk. For some applications, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports,

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from which the engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the prosthesis includes a valve including a collapsible pliant material, configured to assume a closed position during diastole and an open position during systole, and the pliant material includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native aortic valve of a native valve complex of a subject, the native valve complex having exactly two aortic sinuses and two native commissures, the prosthesis including a valve prosthesis support, which includes a support structure including exactly two engagement arms that meet one another at two respective junctures,

wherein the engagement arms are shaped so as define two peak complexes at the two respective junctures, and two trough complexes, each of which is between the peak complexes, and

wherein upon implantation of the prosthesis, each of the engagement arms is at least partially disposed within a respective one of the aortic sinuses, such that each of the peak complexes is disposed distal to and in rotational alignment with a respective one of the native commissures, and each of the trough complexes is disposed at least partially within the respective one of the aortic sinuses.

In an embodiment, the engagement arms are shaped such that each of the peak complexes includes exactly one peak at its respective one of the junctures. In an embodiment, the engagement arms are shaped such that each of the trough complexes includes exactly one trough.

In an embodiment, each of the engagement arms is configured to engage a respective one of the aortic sinuses upon implantation of the prosthesis.

There is still further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

a prosthetic valve including one or more prosthetic leaflets configured to assume a closed position during diastole and an open position during systole; and

a valve prosthesis support, coupled to the prosthetic valve, and configured to engage one or more semilunar sinuses of the native semilunar valve site, such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the semilunar sinuses include respective aortic sinuses, and the valve prosthetic support is configured to engage the one or more aortic sinuses. In an embodiment, the native semilunar valve includes a native pulmonary valve, the semilunar sinuses include respective pulmonary sinuses, and the valve prosthetic support is configured to engage the one or more pulmonary sinuses.

There is yet further provided, in accordance with an embodiment of the present invention, a method for implanting a prosthesis at a native semilunar valve of a native valve complex of a subject, the native valve complex having three semilunar sinuses and three native commissures, the method including:

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providing the prosthesis including a valve prosthesis support, which valve prosthesis support includes a support structure including exactly three engagement arms that meet one another at three respective junctures, and the engagement arms are shaped so as define three peak complexes at the three respective junctures, and three trough complexes, each of which is between two of the peak complexes; and

implanting the prosthesis such that each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses, each of the peak complexes is disposed distal to and in rotational alignment with a respective one of the native commissures, and each of the trough complexes is disposed at least partially within the respective one of the semilunar sinuses.

In an embodiment, the native semilunar valve includes a native aortic valve of the subject, the semilunar sinuses include respective aortic sinuses, and implanting includes implanting the prosthesis such that each of the engagement arms is disposed at least partially within the respective one of the aortic sinuses.

In an embodiment, the native semilunar valve includes a native pulmonary valve of the subject, the semilunar sinuses include respective pulmonary sinuses, and implanting includes implanting the prosthesis such that each of the engagement arms is disposed at least partially within the respective one of the pulmonary sinuses.

In an embodiment, the prosthesis is configured such that, during implantation of the prosthesis, the peak complexes self-align with the respective native commissures.

In an embodiment, implanting includes implanting the prosthesis such that the prosthesis embraces, such as gently embraces, without squeezing, leaflets of the native semilunar valve. In an embodiment, implanting includes implanting the prosthesis such that the valve prosthesis support does not fold over leaflets of the native semilunar valve.

In an embodiment, implanting includes implanting the prosthesis such that the engagement arms touch respective floors of the respective semilunar sinuses.

In an embodiment, implanting includes causing the prosthesis to self-align with respect to the native semilunar valve site by gently rotating the prosthesis.

In an embodiment, the support structure is configured to serve as a distal fixation member, the valve prosthesis support includes a proximal fixation member, and implanting includes implanting the prosthesis such that the proximal fixation member and the engagement arms of the distal fixation member axially sandwich the native valve complex from ventricular and downstream sides thereof, respectively.

In an embodiment, implanting includes:

positioning the distal fixation member in a downstream artery while the distal fixation member is collapsed; expanding the distal fixation member; and

thereafter, positioning the proximal fixation member at least partially on the ventricular side of the native valve complex, the downstream artery selected from the group consisting of: an ascending aorta, and a pulmonary trunk.

In an embodiment, implanting includes:

storing the proximal and distal fixation members in at least one tube selected from the group consisting of: an overtube and a trocar, while the proximal and distal fixation members are collapsed; and

deploying the proximal and distal fixation members from the selected tube such that the proximal and distal fixation members expand.

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In an embodiment, the proximal fixation member includes an inner support structure, the distal fixation member includes an outer support structure that is placed partially over the inner support structure, and implanting includes configuring the inner and outer support structures to one another during the implanting.

There is additionally provided, in accordance with an embodiment of the present invention, a method for implanting a prosthesis at a native aortic valve of a native valve complex of a subject, the native valve complex having exactly two aortic sinuses and two native commissures, the method including:

providing the prosthesis including a valve prosthesis support, which valve prosthesis support includes a support structure including exactly two engagement arms that meet one another at two respective junctures, and the engagement arms are shaped so as define two peak complexes at the two respective junctures, and two trough complexes, each of which is between the peak complexes; and

implanting the prosthesis such that each of the engagement arms is at least partially disposed within a respective one of the aortic sinuses, each of the peak complexes is disposed distal to and in rotational alignment with a respective one of the native commissures, and each of the trough complexes is disposed at least partially within the respective one of the aortic sinuses.

There is still additionally provided, in accordance with an embodiment of the present invention, a method for implanting a prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing the prosthesis including a prosthetic valve including one or more prosthetic leaflets configured to assume a closed position during diastole and an open position during systole, and a valve prosthesis support, coupled to the prosthetic valve; and

implanting the prosthesis such that the valve prosthesis support engages one or more semilunar sinuses of the native semilunar valve site, such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, and implanting the prosthesis includes implanting the prosthesis such that the valve prosthesis support engages the one or more semilunar sinuses of the native aortic valve.

In an embodiment, the native semilunar valve includes a native pulmonary valve, and implanting the prosthesis includes implanting the prosthesis such that the valve prosthesis support engages the one or more semilunar sinuses of the native pulmonary valve.

In an embodiment, implanting the prosthesis includes implanting the prosthesis such that the prosthesis leaflets do not engage the semilunar sinuses.

In an embodiment, implanting the prosthesis includes causing the prosthesis to self-align with respect to the native semilunar valve site by gently rotating the prosthesis.

There is yet additionally provided, in accordance with an embodiment of the present invention, a method, including:

placing a semilunar valve prosthesis at a native semilunar valve site, which prosthesis includes a prosthetic valve including one or more prosthetic leaflets configured to assume a closed position during diastole and an open position during systole; and

engaging a portion of the semilunar valve prosthesis, other than the prosthetic leaflets, with one or more semilunar sinuses of the native semilunar valve site,

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such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of a native semilunar valve of the native semilunar valve site.

In an embodiment, the native semilunar valve site includes a native aortic valve site, the semilunar sinuses include respective aortic sinuses, the semilunar valve prosthesis includes an aortic valve prosthesis, placing includes placing the aortic valve prosthesis at the native aortic valve site, and engaging includes engaging the portion of the aortic valve prosthesis with the one or more aortic sinuses.

In an embodiment, the native semilunar valve site includes a native pulmonary valve site, the semilunar sinuses include respective pulmonary sinuses, the semilunar valve prosthesis includes a pulmonary valve prosthesis, placing includes placing the pulmonary valve prosthesis at the native pulmonary valve site, and engaging includes engaging the portion of the pulmonary valve prosthesis with the one or more pulmonary sinuses.

In an embodiment, engaging includes causing the semilunar valve prosthesis to self-align with respect to the native semilunar valve site by gently rotating the semilunar valve prosthesis.

There is also provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the native valve complex having semilunar sinuses, the prosthesis including a valve prosthesis support, which includes a support structure including at least two engagement arms,

wherein, upon implantation of the prosthesis, each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses, and

wherein a shape of at least one of the engagement arms is generally characterized by a function $z''(r) > 0$, where z is a height of any given point on the at least one engagement arm measured along a longitudinal axis of the prosthesis, and r is a distance from the longitudinal axis to the given point.

For some applications, the shape is generally characterized by the function $z''(r) > 0$.

In an embodiment, the native semilunar valve includes a native aortic valve of the subject, the semilunar sinuses include respective aortic sinuses, and, upon implantation of the prosthesis, each of the engagement arms is disposed at least partially within the respective one of the aortic sinuses.

In an embodiment, the native semilunar valve includes a native pulmonary valve of the subject, the semilunar sinuses include respective pulmonary sinuses, and, upon implantation of the prosthesis, each of the engagement arms is at least partially disposed within the respective one of the pulmonary sinuses.

For some applications, each of the engagement arms includes at least one extension element that extends from the engagement arm, which at least one extension element is configured to engage a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis.

In an embodiment, the support structure includes exactly three engagement arms.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the valve prosthesis support is configured such that, upon implantation of the prosthesis, the valve prosthesis support does not fold over leaflets of the native semilunar valve.

In an embodiment, the support structure is configured to serve as a distal fixation member, the valve prosthesis

support includes a proximal fixation member, and the proximal fixation member and the engagement arms of the distal fixation member are configured to axially sandwich the native valve complex from ventricular and downstream sides thereof, respectively, upon implantation of the prosthesis.

In an embodiment, each of the engagement arms is configured to engage a respective one of the semilunar sinuses upon implantation of the prosthesis.

For some applications, each of the engagement arms is shaped so as to define at least one extension element that extends from the engagement arm, and each of the engagement arms and its respective at least one extension element are configured such that the engagement arm engages, via the at least one extension element, a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis.

For some applications, each of the engagement arms is shaped to define a length, parallel to a longitudinal axis of the prosthesis, between (a) at least one of the junctures and (b) a contact point of one of the engagement arms that meets at the juncture with a sinus floor of the respective one of the semilunar sinuses upon implantation of the prosthesis, which length is greater than 6 mm.

In an embodiment, the prosthesis includes a prosthetic valve including one or more prosthetic leaflets, at least a portion of each of the prosthetic leaflets is configured to assume a closed position during diastole and an open position during systole, and the at least a portion is not directly coupled to any of the engagement arms. For some applications, the prosthetic valve is coupled to the support structure such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve, upon implantation of the prosthesis.

In an embodiment, the engagement arms are configured to touch respective floors of the respective semilunar sinuses, upon implantation of the prosthesis.

In an embodiment, the engagement arms are configured to firmly engage the respective semilunar sinuses, upon implantation of the prosthesis.

There is further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the native valve complex having semilunar sinuses, the prosthesis including a valve prosthesis support, which includes a support structure including at least two engagement arms,

wherein, upon implantation of the prosthesis, each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses, and wherein a shape of at least one of the engagement arms is generally upwardly concave.

There is still further provided, in accordance with an embodiment of the present invention, a method for implanting a prosthesis at a native semilunar valve of a native valve complex of a subject, the native valve complex having semilunar sinuses, the method including:

providing the prosthesis including a valve prosthesis support, which valve prosthesis support includes a support structure including at least two engagement arms, and a shape of at least one of the engagement arms is generally characterized by a function $z''(r) \geq 0$, where z is a height of any given point on the at least one engagement arm measured along a longitudinal axis of the prosthesis, and r is a distance from the longitudinal axis to the given point; and

implanting the prosthesis such that each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses.

In an embodiment, implanting includes implanting the prosthesis such that each of the engagement arms is configured to engage a respective one of the semilunar sinuses.

There is yet further provided, in accordance with an embodiment of the present invention, a method for implanting a prosthesis at a native semilunar valve of a native valve complex of a subject, the native valve complex having semilunar sinuses, the method including:

providing the prosthesis including a valve prosthesis support, which valve prosthesis support includes a support structure including at least two engagement arms, and a shape of at least one of the engagement arms is generally upwardly concave; and

implanting the prosthesis such that each of the engagement arms is at least partially disposed within a respective one of the semilunar sinuses.

There is additionally provided, in accordance with an embodiment of the present invention, a method including: providing a semilunar valve prosthesis; and implanting the prosthesis without using any imaging techniques.

In an embodiment, providing the semilunar valve prosthesis includes providing an aortic valve prosthesis. In an embodiment, providing the semilunar valve prosthesis includes providing a pulmonary valve prosthesis.

In an embodiment, implanting includes: placing the prosthesis at a semilunar valve site; and causing the prosthesis to self-align with respect to the site by gently rotating the prosthesis.

In an embodiment, implanting the prosthesis includes determining a correct rotational disposition of the prosthesis with respect to a semilunar valve site based on tactile feedback.

There is still additionally provided, in accordance with an embodiment of the present invention, a method including: providing a semilunar valve prosthesis; placing the prosthesis in a body of a subject; and determining a correct rotational disposition of the prosthesis with respect to a semilunar valve site based on tactile feedback.

In an embodiment, providing the semilunar valve prosthesis includes providing an aortic valve prosthesis. In an embodiment, providing the semilunar valve prosthesis includes providing a pulmonary valve prosthesis.

In an embodiment, placing the prosthesis includes placing the prosthesis without using any imaging techniques.

There is yet additionally provided, in accordance with an embodiment of the present invention, a method including: placing a semilunar valve prosthesis at a native semilunar valve site; and causing the prosthesis to self-align with respect to the site by gently rotating the valve prosthesis.

In an embodiment, the semilunar valve prosthesis includes an aortic valve prosthesis, the native semilunar valve site includes a native aortic valve site, and placing includes placing the aortic valve prosthesis at the native aortic valve site. In an embodiment, the semilunar valve prosthesis includes a pulmonary valve prosthesis, the native semilunar valve site includes a native pulmonary valve site, and placing includes placing the pulmonary valve prosthesis at the native pulmonary valve site.

In an embodiment, causing the prosthesis to self-align includes moving the prosthesis in an axial direction defined with respect to an axis of a downstream artery, while gently

rotating the prosthesis, the downstream artery selected from the group consisting of: an ascending aorta, and a pulmonary trunk.

In an embodiment, gently rotating the prosthesis includes moving the prosthesis in a proximal direction such that contact of the prosthesis with tissue of the native semilunar valve site causes the rotating.

In an embodiment, placing the prosthesis and causing the prosthesis to self-align include placing the prosthesis and causing the prosthesis to self-align without using any imaging techniques.

In an embodiment, causing the prosthesis to self-align includes verifying that the prosthesis is properly aligned with respect to the semilunar valve site by attempting to rotate the prosthesis with respect to the semilunar valve site.

In an embodiment, the prosthesis is shaped so as to define one or more proximal engagement arms that are configured to be positioned at least partially within respective semilunar sinuses of the native semilunar valve site, and causing the prosthesis to self-align includes causing the engagement arms to self-align with respect to the respective semilunar sinuses.

In an embodiment, gently rotating the prosthesis includes moving the prosthesis in a proximal direction such that contact of one or more of the engagement arms with tissue of the native semilunar valve site causes the rotating.

In an embodiment, causing the prosthesis to self-align includes verifying that the engagement arms are properly placed with respect to the semilunar valve site by attempting to rotate the engagement arms with respect to the semilunar valve site.

There is also provided, in accordance with an embodiment of the present invention, a method, including:

placing a semilunar valve prosthesis at a native semilunar valve site, the prosthesis shaped so as to define one or more proximal engagement arms;

attempting to position the engagement arms at least partially within respective semilunar sinuses of the native semilunar valve site; and

verifying that the engagement arms are properly placed with respect to the semilunar valve site by attempting to rotate the engagement arms with respect to the semilunar valve site.

In an embodiment, the semilunar valve prosthesis includes an aortic valve prosthesis, the native semilunar valve site includes a native aortic valve site, and placing includes placing the aortic valve prosthesis at the native aortic valve site.

In an embodiment, the semilunar valve prosthesis includes a pulmonary valve prosthesis, the native semilunar valve site includes a native pulmonary valve site, and placing includes placing the pulmonary valve prosthesis at the native pulmonary valve site.

There is further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including a support structure, which is configured such that a correct rotational disposition of the prosthesis with respect to the native semilunar valve can be determined based on tactile feedback.

There is still further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the native valve complex having semilunar sinuses and native commissures, the prosthesis including:

a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and shaped so as to define exactly three proximal engagement arms that are configured to be positioned at least partially within respective ones of the semilunar sinuses, and, in combination, to apply, to tissue that defines the semilunar sinuses, a first axial force directed toward a ventricle of the subject; and

a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native semilunar valve, and to apply, to the ventricular side of the native valve complex, a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex.

In an embodiment, the native semilunar valve includes a native aortic valve, and the downstream artery includes the ascending aorta, the semilunar sinuses include respective aortic sinuses, and the distal fixation member is configured to be positioned in the ascending aorta, and the proximal engagement arms are configured to be positioned at least partially within the respective aortic sinuses.

In an embodiment, the native semilunar valve includes a native pulmonary valve, and the downstream artery includes the pulmonary trunk, and the semilunar sinuses include respective pulmonary sinuses, and the distal fixation member is configured to be positioned in the pulmonary trunk, and the proximal engagement arms are configured to be positioned at least partially within the respective pulmonary sinuses.

In an embodiment, the distal and proximal fixation members are configured to couple the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof, upon implantation of the prosthesis.

In an embodiment, the distal fixation member does not press upon the native commissures upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply a radial force of less than 0.5 pounds outwardly against the native semilunar valve. In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole. In an embodiment, the prosthesis is configured to apply the second axial force with a force of at least 1 g during systole.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the distal fixation member is configured to be positioned in the downstream artery during an implantation procedure before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex.

In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the distal fixation member is configured such that it does not push leaflets of the native semilunar valve

towards semilunar sinus floors of the native valve complex upon implantation of the prosthesis.

In an embodiment, each of the proximal engagement arms is shaped so as to define at least one trough that is configured to be positioned at least partially within a respective one of the semilunar sinuses.

In an embodiment, the three engagement arms meet one another at three respective junctures, the engagement arms are shaped so as to define three peak complexes at the three respective junctures, and three trough complexes, each of which is between two of the peak complexes, and upon implantation of the prosthesis, at least a portion of each of the peak complexes is disposed distal to and in rotational alignment with a respective one of the native commissures, and each of the trough complexes is disposed at least partially within the respective one of the semilunar sinuses.

In an embodiment, the engagement arms are configured to be positioned, during an implantation procedure, at least partially within the respective ones of the semilunar sinuses before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex, such that the engagement arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount, the opening being because of force applied by the proximal fixation member to the leaflets.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1.

In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native semilunar valve complex.

In an embodiment, the distal fixation member is configured to elevate leaflets of the native semilunar valve from within the semilunar sinuses upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured to apply the first axial force to respective roots of one or more leaflets of the native valve complex. In an embodiment, the distal fixation member is configured to apply the first axial force to respective transitions between respective semilunar sinus floors and one or more leaflets of the native valve complex.

In an embodiment, the prosthesis is configured to apply the first axial force such that the ratio is greater than 3:1, such as greater than 6:1.

In an embodiment, the prosthesis is configured to apply the second axial force such that a ratio of (a) the second axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1, such as greater than 3:1, e.g., greater than 6:1.

In an embodiment, the prosthesis includes a prosthetic valve configured to assume a closed position during diastole and an open position during systole. In an embodiment, the prosthetic valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the prosthetic valve are configured to define a single flow field through the distal and proximal fixation members and the prosthetic valve. Alternatively, the distal and proximal fixation members and the prosthetic valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the prosthetic valve.

In an embodiment, the prosthetic valve includes one or more prosthetic leaflets, and the prosthetic valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured to apply the first axial force to one or more semilunar sinus floors of the native valve complex.

In an embodiment, the distal fixation member is configured not to apply force to leaflets of the native semilunar valve.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the ventricular side of the native valve complex. For some applications, the at least one barb is configured to pierce the ventricular side of the native valve complex. Alternatively, the at least one barb is configured to protrude into tissue of the ventricular side of the native valve complex, without piercing the tissue. For some applications, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible. For some applications, the distal fixation member is configured to be positioned, during an implantation procedure, in the downstream artery while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex. For some applications, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure.

In an embodiment, the outer support structure is shaped so as to define exactly three distal diverging strut supports, from which respective ones of the proximal engagement arms extend radially outward.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the engagement arms are aligned by rotation with respective ones of the semilunar sinuses.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the strut supports are aligned with respective ones of the native commissures.

In an embodiment, the prosthesis is configured such that the engagement arms self-align themselves by rotation during implantation of the prosthesis at the native valve complex.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply the second axial force.

In an embodiment, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define exactly three distal diverging strut supports,

from which respective ones of the proximal engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the prosthesis includes a valve including a collapsible pliant material, configured to assume a closed position during diastole and an open position during systole, and the pliant material includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is yet further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply, to tissue that defines one or more semilunar sinuses of the native valve complex, a first axial force directed toward a ventricle of the subject; and

a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native semilunar valve, and to apply, to the ventricular side of the native valve complex, a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, the semilunar sinuses include respective aortic sinuses, and the distal fixation member is configured to be positioned in the ascending aorta, and to apply the first axial force to the tissue that defines the one or more aortic sinuses.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, the semilunar sinuses include respective pulmonary sinuses, and the distal fixation member is configured to be positioned in the pulmonary trunk, and to apply the first axial force to the tissue that defines the one or more pulmonary sinuses.

In an embodiment, the distal and proximal fixation members are configured to couple the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof, upon implantation of the prosthesis.

In an embodiment, the distal fixation member does not press upon native valve commissures of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply a radial force of less than 0.5 pounds outwardly against the native semilunar valve. In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole. In an embodiment, the prosthesis is configured to apply the second axial force with a force of at least 1 g during systole.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the distal fixation member is configured to be positioned in the downstream artery during an implantation procedure before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex.

In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the distal fixation member is configured such that it does not push leaflets of the native semilunar valve towards semilunar sinus floors of the native valve complex upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex.

In an embodiment, the distal fixation member is configured to apply the first axial force to respective roots of one or more leaflets of the native valve complex. In an embodiment, the distal fixation member is configured to apply the first axial force to respective transitions between respective semilunar sinus floors and one or more leaflets of the native valve complex.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1, such as greater than 3:1, e.g., greater than 6:1.

In an embodiment, the prosthesis is configured to apply the second axial force such that a ratio of (a) the second axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1, such as greater than 3:1, e.g., greater than 6:1.

In an embodiment, the prosthesis includes a prosthetic valve configured to assume a closed position during diastole and an open position during systole. In an embodiment, the prosthetic valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the prosthetic valve are configured to define a single flow field through the distal and proximal fixation members and the prosthetic valve. Alternatively, the distal and proximal fixation members and the prosthetic valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the prosthetic valve.

In an embodiment, the prosthetic valve includes one or more prosthetic leaflets, and the prosthetic valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured to apply the first axial force to one or more semilunar sinus floors of the native valve complex.

In an embodiment, the distal fixation member is configured not to apply force to leaflets of the native semilunar valve.

In an embodiment, the distal fixation member is shaped so as to define one or more proximal engagement arms that are configured to be positioned at least partially within respective ones of the semilunar sinuses, and, in combination, to apply the first axial force.

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In an embodiment, the distal fixation member is shaped so as to define exactly three proximal engagement arms.

In an embodiment, each of the proximal engagement arms is shaped so as to define at least one trough that is configured to be positioned at least partially within a respective one of the semilunar sinuses.

In an embodiment, the three engagement arms meet one another at three respective junctures, the engagement arms are shaped so as to define three peak complexes at the three respective junctures, and three trough complexes, each of which is between two of the peak complexes, and upon implantation of the prosthesis, at least a portion of each of the peaks is disposed distal to and in rotational alignment with a respective native commissure of the native semilunar valve, and each of the trough complexes is disposed at least partially within the respective one of the semilunar sinuses.

In an embodiment, the distal fixation member is shaped so as to define exactly two proximal engagement arms.

In an embodiment, the engagement arms are configured to be positioned, during an implantation procedure, at least partially within the respective ones of the semilunar sinuses before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex, such that the engagement arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount, the opening being because of force applied by the proximal fixation member to the leaflets.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the ventricular side of the native valve complex. For some applications, the at least one barb is configured to pierce the ventricular side of the native valve complex. Alternatively, the at least one barb is configured to protrude into tissue of the ventricular side of the native valve complex, without piercing the tissue. For some applications, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible. For some applications, the distal fixation member is configured to be positioned, during an implantation procedure, in the downstream artery while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex. For some applications, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the engagement arms are aligned by rotation with respective ones of the semilunar sinuses.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the strut supports are aligned with respective commissures of the native valve complex.

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In an embodiment, the prosthesis is configured such that the engagement arms self-align themselves by rotation during implantation of the prosthesis at the native valve complex.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply the second axial force. For some applications, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the prosthesis includes a valve including a collapsible pliant material, configured to assume a closed position during diastole and an open position during systole, and the pliant material includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is additionally provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

- a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply, to native commissures of the native semilunar valve, a first axial force directed toward a ventricle of the subject, without applying any force to native leaflets of the native semilunar valve, and the distal fixation member is configured to rotationally align with the native semilunar valve; and
- a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native valve complex, and to apply a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof, upon implantation of the prosthesis.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and the distal fixation member is configured to be positioned in the ascending aorta, and to apply the first axial force to the native commissures of the native aortic valve.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and the distal fixation member is configured to be positioned in the pulmonary trunk, and to apply the first axial force to the native commissures of the native pulmonary valve.

In an embodiment, the distal fixation member is configured to rotationally self-align with the native semilunar valve.

In an embodiment, the distal fixation member includes one or more engagement arms that are positioned at least

partially within respective semilunar sinuses of the native valve complex, upon implantation of the prosthesis.

In an embodiment, the engagement arms are configured to apply respective forces to respective floors of the semilunar sinuses, upon implantation of the prosthesis.

In an embodiment, the engagement arms are configured not to apply any force to floors of the semilunar sinuses, upon implantation of the prosthesis.

There is still additionally provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply a first axial force directed toward a ventricle of the subject; and

a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native valve complex, and to apply a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,

wherein the prosthesis is configured to apply a radial force of less than 0.5 pounds outwardly against the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and the distal fixation member is configured to be positioned in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and the distal fixation member is configured to be positioned in the pulmonary trunk.

In an embodiment, the distal fixation member does not press upon native valve commissures of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) the radial force is greater than 1.5:1. In an embodiment, the prosthesis is configured to apply the second axial force such that a ratio of (a) the second axial force to (b) the radial force is greater than 1.5:1. In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole. In an embodiment, the prosthesis is configured to apply the second axial force with a force of at least 1 g during systole.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve. In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the prosthesis includes a valve configured to assume a closed position during diastole and an open position during systole. In an embodiment, the valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a single flow field through the distal and proximal fixation members and the valve. Alternatively, the distal and proximal fixation members and the valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, the valve includes one or more prosthetic leaflets, and the valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the ventricular side of the native valve complex. For some applications, the at least one barb is configured to pierce the ventricular side of the native valve complex. Alternatively, the at least one barb is configured to protrude into tissue of the ventricular side of the native valve complex, without piercing the tissue. For some applications, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible. For some applications, the distal fixation member is configured to be positioned, during an implantation procedure, in the downstream artery while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex. For some applications, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply the second axial force. For some applications, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the prosthesis includes a valve including a collapsible pliant material, configured to assume a closed position during diastole and an open position during systole, and the pliant material includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is yet additionally provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis, which distal fixation member is shaped so as to define exactly three proximal engagement arms;

positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the three proximal engagement arms are positioned at least partially within respective semilunar sinuses of the native valve complex, and, in combination, apply, to tissue that defines the semilunar sinuses, a first axial force directed toward a ventricle of the subject; and

positioning the proximal fixation member at least partially on a ventricular side of the native semilunar valve, such that the proximal fixation member applies, to the ventricular side of the native valve complex, a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

In an embodiment, positioning the distal and proximal fixation members includes positioning the engagement arms at least partially within the respective ones of the semilunar sinuses before positioning the proximal fixation member at least partially on the ventricular side of the native valve complex, such that the engagement arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount, the opening being because of force applied by the proximal fixation member to the leaflets.

There is also provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;

positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies, to a downstream side of the native valve complex, a first axial force directed toward a ventricle of the subject; and

positioning the proximal fixation member at least partially on a ventricular side of the native semilunar valve, such that the proximal fixation member applies, to a ventricular side of the native semilunar valve, a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

In an embodiment, positioning the distal and proximal fixation members includes positioning the distal fixation member in the downstream artery before positioning the proximal fixation member at least partially on the ventricular side of the native semilunar valve.

In an embodiment, the prosthesis includes a prosthetic valve, and positioning the distal fixation member includes positioning the distal fixation member such that the valve assumes a closed position during diastole and an open position during systole.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that it limits an extent of opening of leaflets of the native valve complex.

In an embodiment, positioning the proximal and distal fixation members includes:

collapsing the proximal and distal fixation members; inserting the proximal and distal fixation members, while collapsed, in the ventricle and the downstream artery, respectively; and

expanding the proximal and distal fixation members in the ventricle and the downstream artery, respectively.

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In an embodiment, positioning the distal fixation member includes positioning the distal fixation member in the downstream artery while collapsed, and expanding the distal fixation member before positioning the proximal fixation member at least partially on the ventricular side of the native semilunar valve.

In an embodiment, inserting the proximal and distal fixation members includes storing the proximal and distal fixation members while collapsed in at least one tube selected from the group consisting of: an overtube and a trocar, and expanding the proximal and distal fixation members includes deploying the proximal and distal fixation members from the selected tube.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and inserting the proximal and distal fixation members includes inserting the selected tube through an apex of a heart of the subject, and advancing the selected tube through the ventricle until a distal end of the selected tube passes the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and inserting the proximal and distal fixation members includes inserting the selected tube using a transaortic approach.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, the ventricle includes a right ventricle, and inserting the proximal and distal fixation members includes inserting the selected tube through a free wall of the right ventricle, and advancing the selected tube through the right ventricle past a right ventricular outflow tract of the heart until a distal end of the selected tube passes the native pulmonary valve.

In an embodiment, the proximal fixation member includes an inner support structure, the distal fixation member includes an outer support structure that is placed partially over the inner support structure, and positioning the proximal and distal fixation members includes positioning the inner and outer support structures, respectively.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and positioning the outer support structure includes rotationally aligning the engagement arms with respective ones of the semilunar sinuses.

In an embodiment, positioning the outer support structure includes rotationally aligning the strut supports with respective commissures of the native valve complex.

In an embodiment, aligning the engagement arms and the strut supports includes moving the outer support structure in a proximal direction, such that the engagement arms self-align with the respective ones of the semilunar sinuses.

There is further provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;

positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies, to native commissures of the native semilunar valve, a first axial

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force directed toward a ventricle of the subject, without applying any force to native leaflets of the native semilunar valve;

causing the distal fixation member to rotationally align with the native semilunar valve by gently rotating the valve prosthesis; and

positioning the proximal fixation member at least partially on a ventricular side of the native valve complex, such that the proximal fixation member applies a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, causing the distal fixation member to align includes causing the distal fixation member to rotationally self-align with the native semilunar valve.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

There is still further provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;

positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies a first axial force directed toward a ventricle of the subject; and

positioning the proximal fixation member at least partially on a ventricular side of the native valve complex, such that the proximal fixation member applies a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof, and the prosthesis applies a radial force of less than 0.5 pounds outwardly against the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the

ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

There is yet further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply a first axial force directed toward a ventricle of the subject; and

a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native valve complex, and to apply a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,

wherein the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and the distal fixation member is configured to be positioned in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and the distal fixation member is configured to be positioned in the pulmonary trunk.

In an embodiment, the prosthesis is configured such that the radial force is less than 0.5 pounds. In an embodiment, the distal fixation member does not press upon native valve commissures of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply the first axial force such that the ratio is greater than 3:1, such as greater than 6:1.

In an embodiment, the prosthesis includes a valve configured to assume a closed position during diastole and an open position during systole.

In an embodiment, the valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a single flow field through the distal and proximal fixation members and the valve.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, the valve includes one or more prosthetic leaflets, and the valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the ventricular side of the native valve complex. For some applications, the at least one barb is configured to pierce the ventricular side of the native valve complex. Alternatively, the at least one barb is configured to protrude into tissue of the ventricular side of the native valve complex, without piercing the tissue. For some applications, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible. For some applications, the distal fixation member is configured to be positioned, during an implantation procedure, in the downstream artery while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the ventricular side of the native valve complex. For some applications, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward.

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In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply the second axial force. For some applications, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the prosthesis includes a valve including a collapsible pliant material, configured to assume a closed position during diastole and an open position during systole, and the pliant material includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is additionally provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

- a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply a first axial force directed toward a ventricle of the subject; and

- a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native valve complex, and to apply a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,

wherein the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and the distal fixation member is configured to be positioned in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and the distal fixation member is configured to be positioned in the pulmonary trunk.

In an embodiment, the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) the radial force is greater than 1.5:1. In an embodiment, the prosthesis is configured to apply the second axial force such that a ratio of (a) the second axial force to (b) the radial force is greater than 1.5:1. In an embodiment, the prosthesis is configured such that the radial force is less than 0.5 pounds.

In an embodiment, the distal fixation member does not press upon native valve commissures of the native semilunar valve upon implantation of the prosthesis.

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In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole. In an embodiment, the prosthesis is configured to apply the second axial force with a force of at least 1 g during systole.

In an embodiment, the prosthesis is configured, upon implantation thereof, to embrace, such as gently embrace, without squeezing, leaflets of the native semilunar valve. In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex.

In an embodiment, the prosthesis includes a valve configured to assume a closed position during diastole and an open position during systole. In an embodiment, the valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a single flow field through the distal and proximal fixation members and the valve. For some applications, the distal and proximal fixation members and the valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, the valve includes one or more prosthetic leaflets, and the valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

There is also provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

- providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;

- positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies a first axial force directed toward a ventricle of the subject, such that a ratio of (a) the first axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1; and

- positioning the proximal fixation member at least partially on a ventricular side of the native valve complex, such that the proximal fixation member applies a second axial force directed toward the downstream artery, and application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before

positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

There is still additionally provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;

positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies a first axial force directed toward a ventricle of the subject; and

positioning the proximal fixation member at least partially on a ventricular side of the native valve complex, such that the proximal fixation member applies a second axial force directed toward the downstream artery, and application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,

wherein positioning the distal and proximal fixation members includes positioning the distal and proximal fixation members such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member before positioning the distal fixation member and before positioning the proximal fixation member.

In an embodiment, providing includes coupling the distal fixation member to the proximal fixation member after performing at least one action selected from the group consisting of: positioning the distal fixation member, and positioning the proximal fixation member.

In an embodiment, the distal fixation member and the proximal fixation member which are fabricated as one integrated structure, and providing the distal fixation member coupled to the proximal fixation member includes providing the distal fixation member and the proximal fixation member that are fabricated as one integrated structure.

There is yet additionally provided, in accordance with an embodiment of the present invention, apparatus including a

prosthesis for implantation at a native semilunar valve of a native valve complex of a subject, the prosthesis including:

a distal fixation member, configured to be positioned in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, and to apply a first axial force directed toward a ventricle of the subject; and

a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a ventricular side of the native valve complex, and to apply a second axial force directed toward the downstream artery, such that application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,

wherein the prosthesis is configured, upon implantation thereof, to embrace, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the prosthesis is configured, upon implantation thereof, to gently embrace, without squeezing, the leaflets of the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and the distal fixation member is configured to be positioned in the ascending aorta.

In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and the distal fixation member is configured to be positioned in the pulmonary trunk.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the prosthesis is configured to apply the first axial force such that a ratio of (a) the first axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1. In an embodiment, the prosthesis is configured to apply the second axial force such that a ratio of (a) the second axial force to (b) a radial force applied outwardly by the prosthesis against the native semilunar valve is greater than 1.5:1.

In an embodiment, the prosthesis is configured to apply a radial force of less than 0.5 pounds outwardly against the native semilunar valve.

In an embodiment, the distal fixation member does not press upon native valve commissures of the native semilunar valve upon implantation of the prosthesis.

In an embodiment, the prosthesis is configured to apply the first axial force with a force of at least 40 g during diastole. In an embodiment, the prosthesis is configured to apply the second axial force with a force of at least 1 g during systole.

In an embodiment, the prosthesis is configured such that any radial force applied by the prosthesis outwardly against the native semilunar valve is insufficient by itself to chronically maintain the prosthesis in position with respect to the native valve complex under conditions of normal cardiac motion.

In an embodiment, the distal fixation member is configured such that it does not fold over leaflets of the native semilunar valve upon implantation of the prosthesis. In an embodiment, the prosthesis is configured to less than fully open leaflets of the native valve complex when the prosthesis is implanted at the native valve complex.

In an embodiment, the prosthesis includes a valve configured to assume a closed position during diastole and an open position during systole. In an embodiment, the valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a single flow field through the distal and proximal fixation members and the valve. Alternatively, the distal and proximal fixation members and the valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, the valve includes one or more prosthetic leaflets, and the valve is coupled to the prosthesis such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets of the native semilunar valve upon implantation of the prosthesis.

There is also provided, in accordance with an embodiment of the present invention, apparatus including a valve prosthesis for implantation at a native semilunar valve of a subject, the prosthesis including:

- one or more distal fixation members, which are configured to be coupled without suturing to the native semilunar valve such that the members prevent opening of native leaflets of the native semilunar valve to their maximum diameter; and
- a pliant material coupled to at least one of the distal fixation members, the pliant material having a closed position and an open position.

In an embodiment, the native semilunar valve includes a native aortic valve, and the one or more distal fixation members are configured to be coupled with suturing to the native aortic valve. In an embodiment, the native semilunar valve includes a native pulmonary valve, and the one or more distal fixation members are configured to be coupled with suturing to the native pulmonary valve.

In an embodiment, the one or more distal fixation members are configured to define a maximum extent of opening of the native leaflets.

In an embodiment, the one or more distal fixation members include at least two distal fixation members, and the at least two distal fixation members are configured such that upon implantation of the prosthesis, at least a portion of the native leaflets is positioned between the at least two distal fixation members.

There is further provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a native valve complex of a subject, the method including:

- providing a distal fixation member of the valve prosthesis coupled to a proximal fixation member of the valve prosthesis;
- positioning the distal fixation member in a downstream artery of the subject selected from the group consisting of: an ascending aorta, and a pulmonary trunk, such that the distal fixation member applies a first axial force directed toward a ventricle of the subject; and
- positioning the proximal fixation member at least partially on a ventricular side of the native valve complex, such that the proximal fixation member applies a second axial force directed toward the downstream artery, and application of the first and second forces couples the prosthesis to the native valve complex by axially sandwiching the native valve complex from a downstream side and the ventricular side thereof,
- wherein positioning the distal and proximal fixation members includes positioning the distal and proximal fixa-

tion members such that the valve prosthesis embraces, without squeezing, leaflets of the native semilunar valve.

In an embodiment, the native semilunar valve includes a native aortic valve, the downstream artery includes the ascending aorta, and positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta. In an embodiment, the native semilunar valve includes a native pulmonary valve, the downstream artery includes the pulmonary trunk, and positioning the distal fixation member includes positioning the distal fixation member in the pulmonary trunk.

In an embodiment, positioning the distal and proximal fixation members includes positioning the distal and proximal fixation members such that the valve prosthesis gently embraces, without squeezing, the leaflets of the native semilunar valve.

There is still further provided, in accordance with an embodiment of the present invention, a method for implanting a valve prosthesis at a native semilunar valve of a subject, the method including:

- positioning one or more distal fixation members of the valve prosthesis in a vicinity of the native semilunar valve, and a pliant material coupled to at least one of the distal fixation members has a closed position and an open position; and
- without suturing, coupling the one or more distal fixation members to the native semilunar valve such that the distal fixation members prevent opening of native leaflets of the native semilunar valve to their maximum diameter.

In an embodiment, the native semilunar valve includes a native aortic valve, and positioning includes positioning the one or more distal fixation members in the vicinity of the native aortic valve.

In an embodiment, the native semilunar valve includes a native pulmonary valve, and positioning includes positioning the one or more distal fixation members in the vicinity of the native pulmonary valve.

In an embodiment, positioning the one or more distal fixation members includes positioning the one or more distal fixation members to define a maximum extent of opening of the native leaflets.

In an embodiment, the one or more distal fixation members include at least two distal fixation members, and positioning includes positioning the at least two distal fixation members such that at least a portion of the native leaflets are positioned between the at least two distal fixation members.

There is further provided, in accordance with an embodiment of the present invention, apparatus including a prosthesis for implantation at a stenosed native aortic valve of a native valve complex of a subject, the prosthesis including:

- a distal fixation member, configured to be positioned in an ascending aorta of the subject, and to apply, to an aortic side of the native valve complex, a first axial force directed toward a left ventricle of the subject; and
- a proximal fixation member coupled to the distal fixation member, the proximal fixation member configured to be positioned at least partially on a left-ventricular side of the native aortic valve, and to apply, to a left-ventricular side of the aortic annulus, a second axial force directed toward the ascending aorta, such that application of the first and second forces couples the prosthesis to the native valve complex.

In an embodiment, the distal fixation member is configured to be positioned in the ascending aorta during an

implantation procedure before the proximal fixation member is positioned at least partially on the left-ventricular side of the native aortic valve.

In an embodiment, the distal fixation member is configured such that it does not crimp, fold, or compress leaflets of the native aortic valve upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured such that it does not push leaflets of the native aortic valve towards aortic sinus floors of the native valve complex upon implantation of the prosthesis.

In an embodiment, the prosthesis includes a valve configured to assume a closed position during diastole and an open position during systole.

In an embodiment, the valve includes a collapsible pliant material, configured to assume the open and closed positions.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a single flow field through the distal and proximal fixation members and the valve.

In an embodiment, the distal and proximal fixation members and the valve are configured to define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, the prosthesis is configured to not fully open leaflets of the native valve complex when the prosthesis is implanted at the native aortic valve complex.

In an embodiment, the distal fixation member is configured to be positioned within one or more aortic sinuses of the native valve complex upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured to elevate leaflets of the native aortic valve from within the one or more aortic sinuses upon implantation of the prosthesis.

In an embodiment, the distal fixation member is configured to apply the first axial force to respective roots of one or more leaflets of the native valve complex.

In an embodiment, the distal fixation member is configured to apply the first axial force to respective transitions between respective aortic sinus floors and one or more leaflets of the native valve complex.

In an embodiment, the distal fixation member is configured to apply the first axial force to one or more aortic sinus floors of the native valve complex.

In an embodiment, the distal fixation member is shaped so as to define one or more proximal engagement arms that are configured to be positioned within respective ones of the aortic sinuses, and, in combination, to apply the first axial force.

In an embodiment, the arms are configured to be positioned, during an implantation procedure, within the respective ones of the aortic sinuses before the proximal fixation member is positioned at least partially on the left-ventricular side of the native aortic valve, such that the arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount because of force applied by the proximal fixation member to the leaflets.

In an embodiment, the proximal fixation member is configured to be positioned at least partially in a left ventricle of the subject upon implantation of the prosthesis.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb configured to apply a barb force to the left-ventricular side of the aortic annulus.

In an embodiment, the at least one barb is configured to pierce the left-ventricular side of the aortic annulus.

In an embodiment, the at least one barb is configured to protrude into tissue of the left-ventricular side of the aortic annulus, without piercing the tissue.

In an embodiment, the distal fixation member is shaped so as to define at least one mating barb, and the at least one barb of the proximal fixation member is configured to engage the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, the proximal and distal fixation members are collapsible.

In an embodiment, the distal fixation member is configured to be positioned, during an implantation procedure, in the ascending aorta while collapsed, and to be expanded before the proximal fixation member is positioned at least partially on the left-ventricular side of the native aortic valve.

In an embodiment, the apparatus includes at least one tube selected from the group consisting of: an overtube and a trocar, and the proximal and distal fixation members are configured to be stored in the selected tube while collapsed, and to expand upon being deployed from the selected tube.

In an embodiment, the proximal fixation member includes an inner support structure, and the distal fixation member includes an outer support structure that is placed partially over the inner support structure.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the engagement arms are aligned by rotation with respective ones of aortic sinuses of the native valve complex.

In an embodiment, the prosthesis is configured such that, upon implantation at the native valve complex, the strut supports are aligned with respective commissures of the native valve complex.

In an embodiment, the prosthesis is configured such that the engagement arms self-align themselves by rotation during implantation of the prosthesis at the native valve complex.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, a proximal portion of which is configured to apply the second axial force.

In an embodiment, the prosthesis includes a graft covering that covers at least a portion of the skirt.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, and the skirt extends from the inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and each of the strut supports is positioned over a respective one of the inner struts.

In an embodiment, the engagement arms are positioned over a portion of the skirt.

In an embodiment, the membrane includes a plurality of segments, at least two of which are coupled together by one of the strut supports and its respective one of the inner struts.

There is further provided, in accordance with an embodiment of the invention, apparatus including a valve prosthesis for implantation at a stenosed native aortic valve of a subject, the prosthesis including:

one or more fixation members, which are configured to be coupled without suturing to the native aortic valve such

that the members do not open native leaflets of the native aortic valve to their maximum diameter; and a membrane coupled to at least one of the fixation members, the membrane having a closed position and an open position.

There is still further provided, in accordance with an embodiment of the invention, a method for treating a stenosed native aortic valve of a native valve complex of a subject, the method including:

positioning a distal fixation member of a valve prosthesis in an ascending aorta of the subject, such that the distal fixation member applies, to an aortic side of the native valve complex, a first axial force directed toward a left ventricle of the subject; and

positioning a proximal fixation member of the prosthesis at least partially on a left-ventricular side of the native aortic valve, such that the proximal fixation member applies, to a left-ventricular side of the aortic annulus, a second axial force directed toward the ascending aorta, such that application of the first and second forces couples the prosthesis to the native valve.

In an embodiment, positioning the distal and proximal fixation members includes positioning the distal fixation member in the ascending aorta before positioning the proximal fixation member at least partially on the left-ventricular side of the native aortic valve.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that it does not crimp, fold, or compress leaflets of the native aortic valve.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that it does not push leaflets of the native aortic valve towards aortic sinus floors of the native valve complex.

In an embodiment, the prosthesis includes a valve, and positioning the distal fixation member includes positioning the distal fixation member such that the valve assumes a closed position during diastole and an open position during systole.

In an embodiment, the valve includes a collapsible pliant material, and positioning the distal fixation member includes positioning the distal fixation member such that the pliant material assumes the open and closed positions.

In an embodiment, positioning the distal and proximal fixation members and the valve includes positioning the distal and proximal fixation members and the valve such that the distal and proximal fixation members and the valve define a single flow field through the distal and proximal fixation members and the valve.

In an embodiment, positioning the distal and proximal fixation members and the valve includes positioning the distal and proximal fixation members and the valve such that the distal and proximal fixation members and the valve define a plurality of flow fields through the distal and proximal fixation members and the valve.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that it does not fully open leaflets of the native valve complex.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member within one or more aortic sinuses of the native valve complex.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that it elevates leaflets of the native aortic valve from within the one or more aortic sinuses.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that the

distal fixation member applies the first axial force to respective roots of one or more leaflets of the native valve complex.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that the distal fixation member applies the first axial force to respective transitions between respective aortic sinus floors and one or more leaflets of the native valve complex.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member such that the distal fixation member applies the first axial force to one or more aortic sinus floors of the native valve complex.

In an embodiment, the distal fixation member is shaped so as to define one or more proximal engagement arms, and positioning the distal fixation member includes positioning the engagement arms within respective ones of the aortic sinuses, such that the engagement arms apply the first axial force.

In an embodiment, positioning the arms includes positioning the arms before positioning the proximal fixation member, such that the arms prevent leaflets of the native valve complex from opening more than a predetermined desired amount because of force applied by the proximal fixation member to the leaflets.

In an embodiment, positioning the proximal fixation member includes positioning the proximal fixation member at least partially in a left ventricle of the subject.

In an embodiment, the proximal fixation member is shaped so as to define at least one barb, and positioning the proximal fixation member includes positioning the proximal fixation member applies a barb force to the left-ventricular side of the aortic annulus.

In an embodiment, positioning the proximal fixation member includes positioning the proximal fixation member such that the at least one barb pierces the left-ventricular side of the aortic annulus.

In an embodiment, positioning the proximal fixation member includes positioning the proximal fixation member such that the at least one barb protrudes into tissue of the left-ventricular side of the aortic annulus, without piercing the tissue.

In an embodiment, the distal fixation member is shaped so as to define at least one mating barb, and positioning the proximal and distal fixation members includes engaging the at least one barb by the at least one mating barb, so as to help hold the prosthesis in place.

In an embodiment, positioning the proximal and distal fixation members includes:

collapsing the proximal and distal fixation members; inserting the proximal and distal fixation members, while collapsed, in the left ventricle and the ascending aorta, respectively; and

expanding the proximal and distal fixation members in the left ventricle and the ascending aorta, respectively.

In an embodiment, positioning the distal fixation member includes positioning the distal fixation member in the ascending aorta while collapsed, and expanding the distal fixation member before positioning the proximal fixation member at least partially on the left-ventricular side of the native aortic valve.

In an embodiment, inserting the proximal and distal fixation members includes storing the proximal and distal fixation members while collapsed in at least one tube selected from the group consisting of: an overtube and a trocar, and expanding the proximal and distal fixation members includes deploying the proximal and distal fixation members from the selected tube.

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In an embodiment, inserting the proximal and distal fixation members includes inserting the selected tube through an apex of a heart of the subject, and advancing the selected tube through the left ventricle until a distal end of the selected tube passes the native aortic valve.

In an embodiment, inserting the proximal and distal fixation members includes inserting the selected tube using a transaortic approach.

In an embodiment, the proximal fixation member includes an inner support structure, the distal fixation member includes an outer support structure that is placed partially over the inner support structure, and positioning the proximal and distal fixation members includes positioning the inner and outer support structures, respectively.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, and positioning the outer support structure includes rotationally aligning the engagement arms with respective ones of the aortic sinuses.

In an embodiment, positioning the outer support structure includes rotationally aligning the strut supports with respective commissures of the native valve complex.

In an embodiment, aligning the engagement arms and the strut supports includes moving the outer support structure in a proximal direction, such that the engagement arms self-align with the respective ones of the aortic sinuses.

In an embodiment, the inner support structure is shaped so as to define a bulging proximal skirt, and positioning the inner support structure includes positioning the inner support structure such that a proximal portion of the skirt applies the second axial force.

In an embodiment, the prosthesis includes a graft covering that covers at least a portion of the skirt, and positioning the inner support structure includes positioning the inner support structure including the graft covering.

In an embodiment, the inner support structure is shaped so as to define a plurality of distal diverging inner struts, the skirt extends from the inner struts, and positioning the inner support structure includes positioning the inner support structure that is shaped so as to define the plurality of distal diverging inner struts.

In an embodiment, the outer support structure is shaped so as to define a plurality of distal diverging strut supports, from which a plurality of proximal engagement arms extend radially outward, each of the strut supports is positioned over a respective one of the inner struts, and positioning the outer support structure includes positioning the outer support structure that is shaped so as to define the plurality of distal diverging strut supports.

In an embodiment, the engagement arms are positioned over a portion of the skirt, and positioning the outer support structure includes positioning the outer support structure including the engagement arms positioned over the portion of the skirt.

There is yet further provided, in accordance with an embodiment of the invention, a method for treating a stenosed native aortic valve of a subject, the method including:

positioning one or more fixation members of a valve prosthesis in a vicinity of the native aortic valve, and a membrane coupled to at least one of the fixation members has a closed position and an open position; and

without suturing, coupling the one or more fixation members to the native aortic valve such that the fixation

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members do not open native leaflets of the native aortic valve to their maximum diameter.

In some embodiments of the present invention, a fixation mechanism is provided for implanting a stent-based valve prosthesis for treating a native stenosed valve, such as an aortic valve. The fixation mechanism typically enables accurate positioning of the prosthesis in the native valve orifice in a guided self-aligning procedure, as well as safe and secure deployment and fixation.

In some embodiments of the present invention, the fixation mechanism includes one or more of the following components and/or features:

a distal (i.e., downstream) fixation member, which typically includes a fixation frame. When the valve prosthesis is in a collapsed position, the fixation frame is pressed against a body of the valve prosthesis by insertion into an outer sheath (i.e., an overtube);

the downstream fixation frame is shaped so as to define aortic sinus fixation arms, a number of which is typically equal to the number of aortic sinuses of the native valve;

the arms are configured to flare out laterally, when released from the outer sheath, to an angle with respect to a central axis of the prosthesis. Typically, the angle is precisely predefined by the design of the downstream fixation frame and arms, said angle open in the upstream direction. For some applications, the arms are shaped so as to curve outwards laterally;

upon deployment at the bottom of the aortic sinuses, the downstream fixation arms exert force largely or substantially only in the direction of the left ventricle (i.e., an axial force), and exert little or substantially no force in the radial direction;

the downstream fixation arms engage with the downstream side of the native valve leaflets, but not with the upstream side of the native valve leaflets. As a result: (a) the arms limit the opening motion of the native valve leaflets to the above-mentioned angle (which is typically predefined), and (b) the configuration of the arms enables the sequential entrapment of the native valve leaflets, first, from the downstream side by the fixation arms, and, second, from the upstream side, by a proximal (i.e., upstream) fixation member, thereby sandwiching the leaflets at the above-mentioned angle (which is typically predefined) without crimping, folding over, or bending the native leaflets;

the downstream fixation arms engage with an upstream portion of the valve prosthesis to form a locking mechanism, which, for some applications, includes barbs; and/or

divergent commissural struts which encompass at their distal end an area larger than the native aortic orifice, so that the struts help resist migration of the valve prosthesis in an upstream direction (i.e., towards the left ventricle), and contribute to exerting and enhancing axial force in an upstream direction in a manner that increases with their outward angulation and the downstream (aortic) pressure.

In some embodiments of the present invention, the valve prosthesis is implanted using a transapical implantation procedure. An introducer overtube or trocar is inserted into the left ventricular apex using a Seldinger technique. Through this trocar, a delivery catheter onto which the collapsed valve prosthesis (covered by a sheath) is mounted, is advanced into the ascending aorta. Withdrawal of the sheath causes the fixation arms to flare out laterally to an

angle which is typically predetermined by design, and to open in an upstream direction.

Gentle withdrawal and rotation of the delivery catheter, onto which the prosthesis with the flared-out arms is mounted, causes the arms to slide into the aortic sinuses, until the arms reach the bottom (anatomic inferior portion) of the sinuses. This rotational alignment occurs because the three-dimensional geometry of the downstream fixation frame, including the extended aortic sinus fixation arms, conforms to the three-dimensional geometry of the aortic valve and aortic root. In this position, the fixation arms engage with the downstream side of the native valve leaflets, and not with the upstream side of the native valve leaflets. Such engagement limits the opening motion of the native valve leaflets to the above-mentioned angle (which is typically predefined), so that the native leaflets are not pushed against the coronary arteries upon device release. In addition, such engagement provides the proper conditions for sequentially entrapping the native valve leaflets first from the downstream side (by the fixation arms), and subsequently from the upstream side (by the bottom of the valve prosthesis), thereby sandwiching the leaflets at the angle (which is typically predefined), without crimping, folding over, or bending the native leaflets.

Once the proper position of the arms at the bottom of the aortic sinuses is verified, the correct position for complete device release is automatically achieved. The proper position may be verified, for example, by (a) sensing an elastic resistance in the axial direction, and sensing that the device is rotationally locked in place, and/or (b) using imaging techniques such as fluoroscopy and/or ultrasound. Release of the device from the delivery catheter causes a lower inflow portion of the prosthesis to unfold and press against the upstream side of the native leaflets, thereby engaging with the upstream fixation arms in the aortic sinuses. The upstream fixation arms serve as counterparts to the lower inflow portion of the prosthesis in a mechanism that locks the native leaflets and the surrounding periannular tissue for fixation.

Device migration in the upstream direction (into the left ventricle) is prevented by (a) the aortic sinus fixation arms, which exert axial pressure against the bottom of the sinuses, and (b) the outwardly directed angulation of the longitudinally-oriented commissural struts of the prosthesis. The angulation of the struts not only prevents migration into the left ventricle by itself, but, during systole, also by exerting leverage on the aortic sinus fixation arms, which is a function of the degree of the angle and aortic pressure. Migration of the device in a downstream direction is prevented by the inflow part of the device pressing against the periannular tissue surrounding the upstream side of the valve leaflets, and by the inflow part of the device engaging with the fixation arms in a locking mechanism, which, for some applications, includes the use of barbs placed at the inflow section of the device in an upstream direction against the fixation arms.

In other embodiments of the present invention, the valve prosthesis is implanted using another implantation technique, such as an antegrade transseptal technique, or a retrograde endovascular-percutaneous technique.

The present invention will be more fully understood from the following detailed description of embodiments thereof, taken together with the drawings, in which:

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 is a schematic illustration of a fully-assembled valve prosthesis, in accordance with an embodiment of the present invention;

FIG. 2A is a schematic illustration of a collapsible outer support structure of the prosthesis of FIG. 1 prior to assembly with an inner support structure of the prosthesis, in accordance with an embodiment of the present invention;

FIG. 2B is a schematic illustration of the collapsible inner support structure prior to assembly with the outer support structure of the prosthesis of FIG. 1, in accordance with an embodiment of the present invention;

FIGS. 2C and 2D are schematic illustrations of alternative configurations of a portion of the prosthesis of FIG. 1, in accordance with respective embodiments of the present invention;

FIG. 2E is a schematic illustration of another configuration of a collapsible outer support structure of the prosthesis of FIG. 1 prior to assembly with an inner support structure of the prosthesis, in accordance with an embodiment of the present invention;

FIGS. 3A-E are schematic illustrations of additional configurations of the outer support structure of FIG. 2A, in accordance with respective embodiments of the present invention;

FIG. 3F is a schematic illustration of an additional configuration of the outer support structure of FIG. 2A, in accordance with an embodiment of the present invention;

FIG. 3G is a schematic illustration of a fully-assembled valve prosthesis that includes inner engagement arms of the configuration of FIG. 3F, in accordance with an embodiment of the present invention;

FIGS. 4A-C are schematic illustrations of configurations for coupling a pliant material to inner struts of the inner support structure of FIG. 2B and strut supports of the outer support structure of FIG. 2A, in accordance with respective embodiments of the present invention;

FIGS. 4D and 4E are side-view schematic illustrations of configurations for coupling the pliant material of FIGS. 4A-C to a graft covering, in accordance with respective embodiments of the present invention;

FIGS. 5A-C, 6A-B, 7A-E, and 8A illustrate apparatus and a method for implanting the valve prosthesis of FIG. 1 in a native stenosed valve of a heart, in accordance with respective embodiments of the present invention;

FIGS. 8B-C illustrate the prosthesis of FIG. 1 in situ, in accordance with respective embodiments of the present invention;

FIGS. 9A-G schematically illustrate a transaortic approach for implanting the valve prosthesis of FIG. 1, in accordance with an embodiment of the present invention;

FIGS. 10A and 10B show the valve prosthesis of FIG. 1 in open (systolic) and closed (diastolic) positions, respectively, in accordance with an embodiment of the present invention;

FIGS. 11A-D illustrate several configurations for axially coupling the valve prosthesis of FIG. 1 to the aortic annulus, in accordance with respective embodiments of the present invention;

FIGS. 12A-G illustrate a holding device for holding the valve prosthesis of FIG. 1 prior to the implantation of the prosthesis, in accordance with an embodiment of the present invention;

FIGS. 13A-D illustrate the loading of the valve prosthesis of FIG. 1 into a tube from the holding device of FIGS. 12A-G, in accordance with an embodiment of the present invention;

FIG. 14 is a schematic illustration of a valve prosthesis placed in a pulmonary valve, in accordance with an embodiment of the present invention;

FIG. 15 is a schematic anatomical illustration showing the location of a native valve complex, in accordance with an embodiment of the present invention;

FIGS. 16A-H schematically illustrate another transapical technique for implanting the prosthesis of FIG. 1, in accordance with an embodiment of the present invention;

FIG. 17 is a schematic illustration showing a shape of engagement arms of an outer support structure of the prosthesis of FIG. 1, in accordance with an embodiment of the present invention;

FIG. 18 is a schematic illustration of an assembled valve prosthesis support structure, in accordance with an alternative embodiment presented herein;

FIG. 19A is a schematic illustration of a collapsible outer support structure of the valve prosthesis support structure of FIG. 18, prior to assembly with the inner support structure of FIG. 19B, in accordance with an embodiment presented herein; and

FIG. 19B is a schematic illustration of a collapsible inner support structure of the valve prosthesis support structure of FIG. 18, prior to assembly with the outer support structure of FIG. 19A, in accordance with an embodiment of the present invention.

DETAILED DESCRIPTION OF THE EMBODIMENTS

FIG. 1 is a schematic illustration of a fully-assembled valve prosthesis 10, in accordance with an embodiment of the present invention. Valve prosthesis 10 comprises a collapsible inner support structure 12 that serves as a proximal fixation member, and a collapsible outer support structure 14 that serves as a distal fixation member. Outer and inner support structures 14 and 12 may be initially formed separately and then joined together, as shown, or may be formed as one integrated structure, i.e., not formed separately and then joined together. For some applications, outer and inner support structures 14 and 12 are joined together prior to implantation of prosthesis 10 (during a manufacturing process, or by a healthcare worker prior to implantation), while for other applications, the outer and inner support structures are coupled to one another during an implantation procedure. For some applications, outer support structure 14 is constructed from a plurality of separate pieces, which are joined to inner support structure 12 using standard manufacturing means, such as welding, gluing, or suturing (configuration not shown), such that the functionality of outer support structure 14 is attained.

Valve prosthesis 10 is configured to be placed in a native diseased valve of a subject, such as a native stenotic aortic or pulmonary valve, using a minimally-invasive approach, such as a beating heart transapical procedure, such as described hereinbelow with reference to FIGS. 5A-8A or with reference to FIGS. 16A-H, or a retrograde transaortic procedure, such as described hereinbelow with reference to FIGS. 9A-G. As used in the present application, including in the claims, a “native semilunar valve” is to be understood as including: (a) native semilunar valves that include their native leaflets, and (b) native semilunar valves, the native leaflets of which have been surgically excised or are otherwise absent.

Reference is made to FIG. 2A, which is a schematic illustration of collapsible outer support structure 14 prior to assembly with inner support structure 12, in accordance with an embodiment of the present invention. Outer support structure 14 is shaped so as to define a plurality of distal diverging strut supports 20, from which a plurality of

proximal engagement arms 22 extend radially outward in a proximal direction. Typically, the engagement arms have a shape that is generally upwardly concave, such as described hereinbelow with reference to FIG. 17.

Although three strut supports 20 and engagement arms 22 are shown in the figures, for some applications valve prosthesis 10 comprises fewer or more supports and/or arms, such as two supports and two arms. It is noted that approximately 90% of humans have exactly three aortic sinuses. The three supports and/or arms provided in most embodiments correspond to these three aortic sinuses. For implantation in the approximately 10% of patients that have exactly two aortic sinuses, prosthesis 10 typically includes exactly two supports and/or arms.

Engagement arms 22 are typically configured to be at least partially disposed within aortic sinuses of the subject, and, for some applications, to engage and/or rest against floors of the aortic sinuses, and to apply an axial force directed toward a left ventricle of the subject. Engagement arms 22 meet one another at respective junctures 24. For applications in which each of engagement arms 22 is fabricated as a separate piece, the engagement arms are mechanically engaged to one another where they meet at respective junctures 24. For some applications, engagement arms 22 meet one another without actually touching one another, and instead meet via an area defined at each respective juncture 24. Typically, the engagement arms are configured to define respective peaks at junctures 24 (or peak complexes, as described hereinbelow with reference to FIG. 3E), and respective troughs 26 between each two of the peaks (or trough complexes, as described hereinbelow with reference to FIG. 3E).

Outer support structure 14 comprises a suitable material that allows mechanical deformations associated with crimping and expansion of valve prosthesis 10, such as, but not limited to, nitinol or a stainless steel alloy (e.g., AISI 316). Outer support structure 14 is fabricated from a single piece or from a plurality of parts that are coupled together (e.g., by suturing). For some applications, placement of engagement arms 22 within the aortic sinuses prevents “device migration,” i.e., undesired retrograde movement of valve prosthesis 10 that may result from fluid forces applied to the valve. For some applications, engagement arms 22 are coated with a flexible material (e.g., polyester, biocompatible, synthetic, and/or pericardium).

Strut supports 20 and engagement arms 22 may be formed as one integrated structure (as shown), or, alternatively, may be initially formed separately and then joined to one another. For example, the strut support and arms may be mechanically interlocked or sutured together, or coupled by other means. Typically, the strut support and arms are joined prior to implantation.

Reference is made to FIG. 2B, which is a schematic illustration of collapsible inner support structure 12 prior to assembly with outer support structure 14, in accordance with an embodiment of the present invention. For some applications, inner support structure 12 is shaped so as to define a plurality of distal diverging inner struts 30, and a bulging proximal skirt 32 that extends from the struts. A proximal portion 34 of proximal skirt 32 is configured to engage a left ventricular outflow tract (LVOT) of the subject and/or periannular tissue at the top of the left ventricle. A relatively narrow throat section 36 of proximal skirt 32 is configured to be positioned at a valvular annulus of the subject, and to engage the native valve leaflets. Inner support structure 12 comprises, for example, nitinol, a stainless steel alloy, another metal, or another biocompatible material.

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Reference is again made to FIG. 1. Inner and outer support structures 12 and 14 are assembled together by placing outer support structure 14 over inner support structure 12, such that outer strut supports 20 are aligned with, and typically support, respective inner struts 30, and engagement arms 22 are placed over a portion of proximal skirt 32. Inner struts 30 and outer strut supports 20 together function as commissural posts. Typically, such assembly is performed prior to implantation of prosthesis 10, such as during manufacture of the prosthesis; alternatively, such assembly is performed in vivo during an implantation procedure, or prior to implantation by a healthcare worker.

Valve prosthesis 10 typically comprises a prosthetic distal valve 104, which typically comprises a pliant material 105 coupled to strut supports 20 and/or inner struts 30. Pliant material 105 of valve 104 is configured to collapse inwardly (i.e., towards a longitudinal axis of valve prosthesis 10) during diastole, in order to inhibit retrograde blood flow, and to open outwardly during systole, to allow blood flow through the prosthesis. For some applications, when in an open position, valve 104 assumes a diverging shape that causes blood to flow therethrough with pressure recovery at a distal outlet of the valve, for example using techniques described in one or more of the above-mentioned patent application publications to Schwammenthal et al. For other applications, the shape of the valve does not cause such pressure recovery. For example, an angle between the pliant material 105 and a central longitudinal axis of prosthesis 10 may be too great to cause pressure recovery. In this latter case, the large angle may serve exclusively, or at least in part, to help provide axial fixation of prosthesis 10 to the native valve complex. Regardless of whether pressure recovery is achieved, the angle between pliant material 105 and the central longitudinal axis of prosthesis 10 typically inhibits migration of the device in an upstream direction.

Pliant material 105 comprises a flexible supple material, such as an inert biological material, e.g., pericardium sheet or any medically safe elastomer, such as, but not limited to, polyester, polymer, a metallic material/alloy, polyurethane, latex, or synthetic rubber. For some applications, pliant material 105 is coupled to strut supports 20 and/or inner struts 30 by sewing, such as described hereinbelow with reference to FIG. 4. For example, pliant material 105 may be sewn onto outer diverging strut supports 20. Valve 104 comprises a single piece or multiple pieces of pliant material 105 (e.g., leaflets) joined together to give a desired shape, typically a distally diverging shape. For some applications, the pliant material and support structures are coupled to one another in a single-step procedure (e.g., by sewing all the pieces together); alternatively, the pliant material and support structures are coupled to one another in a plurality of sequential steps.

Typically, valve prosthesis 10 further comprises a graft covering 106 which is coupled to proximal skirt 32, such as by sewing the covering within the skirt (configuration shown in FIG. 1) or around the skirt (configuration not shown). Inner support structure 12 thus defines a central structured body for flow passage that proximally terminates in a flared inlet (proximal skirt 32) that is configured to be seated within an LVOT immediately below an aortic annulus/aortic valve. For some applications, graft covering 106 is coupled at one or more sites to pliant material 105.

FIGS. 2C and 2D are schematic illustrations of alternative configurations of a portion of valve prosthesis 10, in accordance with respective embodiments of the present invention. In these configurations, inner support structure 12 and outer support structure 14 are replaced by an element 38, which is

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shaped so as to define first and second portions 40 and 42. First portions 40 serve as support structures, each of which functionally corresponds to a pair of strut support 20 and inner strut 30, described hereinabove with reference to FIGS. 2A and 2B. Pliant material 105 is coupled to support structures 40. Second portions 42 are bent in a proximal direction, such that proximal portions of the second portions define respective engagement arms 22.

In the configuration shown in FIG. 2C, two second portions 42 extend from the distal end of each first portion 40. In the configuration shown in FIG. 2D, element 38 is shaped so as to define two shoulders 44 that extend laterally from each first portion 40. A single second portion 42 extends from each of shoulders 44.

Reference is again made to FIG. 1. In an embodiment of the present invention, inner support structure 12 is shaped so as to define one or more barbs 120, which are configured to pierce or protrude into the ventricular side of the aortic annulus, as described hereinbelow with reference to FIGS. 7A-E. For some applications, one or more of inner struts 30 is shaped so as to define a respective barb, while for other applications, another element of valve prosthesis 10 is shaped so as to define the one or more barbs, such as proximal skirt 32. For some applications, barbs 120 are oriented parallel to a longitudinal axis of valve prosthesis 10, while for other applications, barbs 120 are oriented to form an angle with respect to the longitudinal axis, such as between about -20 degrees (i.e., slanted towards a central axis of the native valve) and about +89 degrees (i.e., slanted away from the central axis of the native valve), such as between about -5 and about +30 degrees. For some applications, barbs 120 are set at the desired angle by heat-setting.

Reference is made to FIG. 2E, which is a schematic illustration of another configuration of collapsible outer support structure 14 prior to assembly with inner support structure 12, in accordance with an embodiment of the present invention. Inter-strut support elements 17 are coupled between adjacent ones of distal diverging strut supports 20, and typically serve to help maintain a desired distance between each of strut supports 20. For example, if a force is applied that would bring closer or separate two of the strut supports, the inter-strut support element between the strut supports would tend to reduce such a deformation. For some applications, one or more of support elements 17 is shaped so as to define a kink or curved section 19, which deforms slightly in response to force applied to element 17.

Reference is made to FIGS. 3A-E, which are schematic illustrations of additional configurations of outer support structure 14, in accordance with respective embodiments of the present invention. In the configurations shown in FIGS. 3A-B, outer support structure 14 is shaped so as to define one or more native valve support elements 122. These support elements apply pressure to an outer (downstream) surface of the native valve when engagement arms 22 are positioned in the aortic sinuses, so as to hold the native leaflets in place against proximal skirt 32. In the configuration shown in FIG. 3A, the area defined by engagement arms 22 and support elements 122 is open, while in the configuration shown in FIG. 3B, a covering 124 is provided in this area. The covering generally may help capture calcific, thrombotic, or other material which might be dislodged from the native valve or the surrounding tissue, and may comprise, for example, polyester. In the configuration shown in FIG. 3C, covering 124 is provided without support elements 122.

In the configuration shown in FIG. 3D, each of engagement arms 22 comprises or is shaped so as to define at least

one extension element **23** that extends from the engagement arm. The engagement arms and extension elements are configured such that the engagement arms engage and/or rest against the floors of the aortic sinuses via the extension elements. For some applications, such as shown in FIG. 3D, exactly one extension element **23** extends from each of engagement arms **22**, while for other applications, more than one extension element **23** extends from each engagement arm (configuration not shown). Although engagement arms **22** are shown in FIG. 3D as curving down toward the sinus floors, for some applications the engagement arms are shaped so as to remain above the native commissures (for example, the engagement arms collectively may be annular in shape), or to curve down less than is shown in FIG. 3D.

In the configuration shown in FIG. 3E, each of engagement arms **22** is shaped so as to define a plurality of troughs **25** and local peaks **27**, rather than a single trough **26**, as shown in FIG. 2A. In addition, each of engagement arms **22** is shaped so as to define a plurality of peaks **29** and local troughs **31**, rather than a single peak at each of junctures **24**, as shown in FIG. 2A. (Outer support structure **14** may include both, only one of, or neither of the features described in the preceding two sentences.) As used in the present application, including in the claims, a “trough complex” means a portion of an engagement arm that extends downwards between respective “peak complexes.” Each “trough complex” includes *n* local troughs **25** and *n*−1 local peaks **29**, where *n* is greater than or equal to one. Each “peak complex” includes *m* local peaks **29** and *m*−1 local troughs **31**, where *m* is greater than or equal to one. It is noted that the portion of a peak complex that is at a juncture may define a local trough (configuration not shown). In addition, although the peak and trough complexes shown in FIG. 3E are generally symmetrical, non-symmetrical arrangements are also within the scope of the present invention.

For some applications, respective extension elements **23**, described hereinabove with reference to FIG. 3D, extend from one or more of the troughs of a trough complex, and/or from elsewhere along the trough complex.

FIG. 3F is a schematic illustration of an additional configuration of outer support structure **14**, in accordance with an embodiment of the present invention. In this embodiment, outer support structure **14**, in addition to defining proximal engagement arms **22**, is shaped so as to define a plurality of inner engagement arms **33**. The inner engagement arms are configured to pass through the valvular annulus. Typically, troughs **35** of inner engagement arms **33** are configured to engage the LVOT and/or periannular tissue at the top of the left ventricle. For some applications, each of inner engagement arms **33** is shaped so as to define one or more barbs **37**, which are configured to pierce or protrude into the ventricular side of the aortic annulus. Typically, during an implantation procedure, inner engagement arms **33** are released from an overtube, trocar, or catheter prior to the release of proximal skirt **32** therefrom, such as described hereinbelow with reference to FIGS. 7A-C, 9A-G, and 16A-H. The fixation provided by inner engagement arms **33** holds prosthesis **10** in place until the implantation procedure is complete, such that blood flow against skirt **32** does not dislodge the prosthesis during the implantation procedure.

FIG. 3G is a schematic illustration of a fully-assembled valve prosthesis that includes inner engagement arms **33** of FIG. 3F, in accordance with an embodiment of the present invention. FIG. 7E, described hereinbelow, shows prosthesis **10** in situ having the configuration shown in FIG. 3F.

For some applications, the features shown in one or more of FIGS. 2A-B and 3A-G are combined. For example, valve

support elements **122** and/or covering **124** may be provided for arms **22** of FIG. 3E. Other such combinations of features are within the scope of the present invention.

Reference is now made to FIGS. 4A-C, which are schematic illustrations of configurations for coupling pliant material **105** to inner struts **30** of inner support structure **12** and to strut supports **20** of outer support structure **14**, in accordance with respective embodiments of the present invention.

In the configuration shown in FIG. 4A, valve **104** comprises a plurality of segments of pliant material **105**, pairs of which are coupled together at respective interfaces between one of inner struts **30** and one of strut supports **20**. Inner strut **30** is shaped so as to define an elongated slit **130**. During manufacture of valve prosthesis **10**, edges of two pieces of pliant material **105** are inserted through slit **130** such that a portion of each of the pieces of pliant material is sandwiched between inner strut **30** and strut support **20**. The inner strut and strut support are tightly coupled together, such as by passing one or more sutures **132** through holes **134** defined by inner strut **30** and strut support **20**. Sutures **132** typically couple the strut and strut support together such that pliant material **105** is supported on both sides thereof, thereby forming a strain relief which reduces stresses on the leaflets of valve **104** at the sutures. The relatively large surface areas of inner strut **30** and strut support **20** distribute the stress applied at pliant material **105**, so that this stress is not applied primarily around holes **134**. Typically, the edges of slit **130** are rounded in order to avoid damage to pliant material **105**.

In the configuration shown in FIGS. 4B-C, portions **136** of graft covering **106** (including, optionally, pericardium or any suitable supple synthetic or biological material) are inserted through slit **130**, between the edges of the slit and the two pieces of pliant material. The portions of the graft covering reduce friction between the pliant material and inner strut **30**. As can be seen in FIG. 4C, portions **136** of graft covering **106** are typically integral with the rest of graft covering **106** (which is sewn to skirt **32**). Graft covering **106** (including, optionally, pericardium or any suitable supple synthetic or biological material) is thus shaped so as to define distally protruding portions **136**.

FIGS. 4D and 4E are side-view schematic illustrations of two configurations for coupling pliant material **105** to graft covering **106**, and reducing leaflet stress during valve opening (FIG. 4D) or valve closure (FIG. 4E), in accordance with respective embodiments of the present invention. In both of these configurations, graft covering **106** is sewn to a cord **107**, such that a portion of pliant material **105** is held between the cord and the graft covering. Cord **107** passes through a hole **108** (FIG. 4C) passing through or near one of the commissural posts (configuration not shown).

Reference is now made to FIGS. 5A-8A, which illustrate apparatus and a method for implanting valve prosthesis **10** in a native stenosed valve **140** of a heart **142**, in accordance with respective embodiments of the present invention.

FIGS. 5A-C illustrate an overtube or trocar **150** and the initial steps of the implantation method, in accordance with respective embodiments of the present invention. Overtube or trocar **150** is placed over a dilator **154**. As shown in FIG. 5A, overtube or trocar **150** is typically inserted through an apex **156** of heart **142**, and advanced into a left ventricle **157** where its motion is terminated, or through left ventricle **157** until the distal end of dilator **154** passes native aortic valve leaflets **158**. For example, apex **156** may be punctured using a standard Seldinger technique, and a guidewire may be advanced into an ascending aorta **160**. Optionally, native

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aortic valve **140** is partially dilated to about 15-20 mm (e.g., about 16 mm), typically using a standard valvuloplasty balloon catheter. (In contrast, full dilation would be achieved utilizing dilation of 20 mm or more.) Overtube or trocar **150** is advanced into the ascending aorta. Overtube or trocar **150** is pushed beyond aortic valve **140** such that the distal end of overtube or trocar **150** is located above the highest point of native aortic valve **140**. Dilator **154** is removed while overtube or trocar **150** remains in place with its distal end located above aortic valve **140**, as shown in FIG. 5B. It is to be understood that the procedure may be modified so that overtube or trocar **150** is placed within the left ventricle and remains within the left ventricle throughout the entire implantation procedure. Valve prosthesis **10** is advanced through the distal end of overtube or trocar **150** into ascending aorta **160** distal to native valve leaflets **158**, as shown in FIG. 5C. Typically, to facilitate this advancement, prior to the implantation procedure valve prosthesis **10** is loaded into a delivery tube **202**, such as described hereinbelow with reference to FIGS. 12A-13D. During the implantation procedure, delivery tube **202** is advanced through overtube or trocar **150**, thereby advancing the valve prosthesis through the overtube or trocar.

FIGS. 6A-B show an implantation of valve prosthesis **10** in ascending aorta **160**, in accordance with an embodiment of the present invention. As mentioned above with reference to FIGS. 5A-C, the distal end of overtube or trocar **150** is positioned past native valve leaflets **158**. The distal end of valve prosthesis **10** is advanced out of overtube or trocar **150** until engagement arms **22** exit overtube or trocar **150** and snap or spring open, as shown in FIG. 6A. Overtube or trocar **150** is gently pulled back until engagement arms **22** are brought into aortic sinuses **164**. For some applications, overtube or trocar **150** and/or valve prosthesis **10** are gently rotated as indicated by arrows **166** in order to align engagement arms **22** with respective aortic sinuses **164**. Although not typically necessary, fluoroscopic, ultrasound, or other surgical imaging techniques may be used to aid in this positioning. Overtube or trocar **150** and valve prosthesis **10** are pulled back slightly, such that engagement arms **22** are positioned within respective aortic sinuses **164**, as shown in FIG. 6B. (Although engagement arms **22** are shown in FIG. 6B as being in contact with the sinus floors, for some applications the engagement arms do not come in contact with the sinus floors, such as described hereinbelow with reference to FIG. 7B.) Typically, valve prosthesis **10** is configured such that when engagement arms **22** are placed properly within aortic sinuses **164**, outer strut supports **20** are aligned with commissures **170** (see, for example, FIG. 8A), thus preventing any possible obstruction of coronary ostia **116** by valve prosthesis **10**. At this point in the implantation procedure, the distal end of valve prosthesis **10** is free of overtube or trocar **150**, and the proximal end of prosthesis **10** remains in overtube or trocar **150**.

For some applications, the use of imaging techniques is not necessary. The careful pulling back of valve prosthesis **10**, without application of excessive force, generally causes each of engagement arms **22** to automatically self-align with a respective aortic sinus **164**, because outer support structure **14**, particularly engagement arms **22**, generally matches the three-dimensional shape of aortic valve **140**. If one of engagement arms **22** comes in contact with a commissure **170** during the careful pulling back of the prosthesis, the arm slides down the slope of the leaflet into the aortic sinus. Typically, arms **22** are evenly distributed around valve prosthesis **10** with a separation of 120 degrees between arms, such that all three arms naturally fall into place in

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respective sinuses upon even just one of the engagement arms achieving proper alignment with a sinus. This natural alignment generally occurs even if the sinuses themselves are not perfectly distributed at 120 degrees from one another.

This alignment process generally ensures positioning of the prosthetic leaflets within the aortic sinuses, thus exposing the prosthetic leaflets to natural blood vortex formation in the aortic sinuses, which contributes to early closure of the prosthetic leaflets, thus reducing closing volume (i.e., leakage through the prosthetic leaflets before fully closing), as well as promoting low-impact closure of the prosthetic leaflets, which typically reduces leaflet wear.

For some applications, a correct rotational disposition of the prosthesis with respect to the aortic valve site is determined by the surgeon based on tactile feedback.

Reference is now made to FIGS. 7A-E, which illustrate valve prosthesis **10** in situ upon completion of the implantation procedure, in accordance with respective embodiments of the present invention. After valve prosthesis **10** is placed properly within native stenosed valve **140**, as described hereinabove with reference to FIGS. 5A-6B, the proximal end of valve prosthesis **10** is released from overtube or trocar **150**, by withdrawing overtube or trocar **150**. Proximal skirt **32** snaps or springs open to at least partially engage, with its proximal portion **34**, the left-ventricular side of native valve **140**, including at least a portion of an inner surface of an LVOT **180**. As a result, valve prosthesis **10** forms an axial engagement system above and below native valve annulus **182** of native valve **140**, which axially sandwiches a native valve complex (as defined hereinbelow with reference to FIG. 15) from the aortic and left-ventricular sides thereof. Native valve leaflets **158** are captured between proximal skirt **32** and engagement arms **22**, typically without applying force along the longitudinal axis of the leaflets, in order to avoid shortening of the length of the leaflets, or forced bending, crimping, or folding over of the leaflets. For some applications, barbs **120**, if provided, pierce aortic annulus **182** on the left-ventricular side of native valve **140**, while for other applications, the barbs are blunt, in which case they generally protrude into the tissue of the aortic annulus, without piercing the tissue. For some applications, support structure **14** is configured to elevate native valve leaflets **158** from within the aortic sinuses.

In the embodiment shown in FIG. 7A, upon the completion of the implantation of prosthesis **10**, engagement arms **22** are positioned within aortic sinuses **164**, such that the ends of the engagement arms touch the floors of the sinuses. Although the ends of the engagement arms are shown touching approximately the radial center of the floors of the sinuses, for some applications, the ends of the engagement arms touch the floors further from leaflets **158** or closer to the leaflets, or touch the body of the leaflets, the roots of the leaflets, or the transition between the sinuses and the leaflet roots. Alternatively, the engagement arms are shorter, such as shown in FIG. 7B, such that they do not reach the floors of the sinuses. Further alternatively, for some applications prosthesis **10** does not comprise arms **22**, as shown in FIG. 7C.

In the embodiment shown in FIG. 7D, prosthesis **10** has been implanted after the native valve leaflets have been excised, in accordance with an embodiment of the present invention.

The embodiment illustrated in FIG. 7E shows valve prosthesis **10** in situ having the configuration of outer support structure **14** described hereinabove with reference to FIG. 3F.

For some applications, barbs **120** are coated or otherwise provided with a surface property for enhancing their attachment to tissue of aortic annulus **182**. Graft covering **106** of proximal skirt **32** also helps prevent regurgitation and device migration.

For some applications, the positioning of arms **22** prior to the opening of proximal skirt **32** prevents native valve leaflets **158** from opening more than a predetermined desired amount. The support provided by arms **22** to the valve leaflets limits the subsequent opening of the leaflets by the proximal skirt. The desired amount of opening is determined at least in part by the angle between arms **22** and a central longitudinal axis of the prosthesis (shown, for example, as angle θ in FIG. 7A). Typically, the angle is between about 1 and about 89 degrees, such as between about 10 and about 60 degrees, such as 25 degrees, or between about 25 and about 65 degrees. Typically, the angle is predetermined. For some applications, the fixation members of prosthesis **10** are configured to prevent opening of the native leaflets to their maximum diameter.

Reference is again made to FIG. 7A. For some applications, prosthetic distal valve **104** is coupled to strut supports **20** and/or inner struts **30** of prosthesis **10** (see, for example, FIG. 1), such that at least 50% of an axial length of the prosthetic leaflets is distal to native valve leaflets **158**. In other words, if prosthetic distal valve **104** has an axial length **L1**, a portion **L2** of length **L1** that is distal to leaflets **158** is greater than a portion **L3** of length **L1** that is proximal to leaflets **158**.

FIG. 8A shows valve prosthesis **10** in situ upon completion of the implantation procedure, as viewed from ascending aorta **160**, upon placement of engagement arms **22** within respective aortic sinuses **164**, in accordance with an embodiment of the present invention. In this embodiment, engagement arms **22** are positioned within aortic sinuses **164**, such that the ends of the engagement arms touch the floors of the sinuses, for example as described hereinabove with reference to FIG. 7A.

FIG. 8B shows valve prosthesis **10** in situ upon completion of the implantation procedure, in accordance with an embodiment of the present invention. In this embodiment, junctures **24** between pairs of engagement arms **22** ride above respective native commissures **170**, without impinging on the commissures (i.e., touching or pushing the commissures). In other words, there is a gap between each of junctures **24** and its respective native commissure **170**. Engagement arms **22** are positioned within aortic sinuses **164**, such that the ends of the engagement arms touch the floors of the sinuses. In this embodiment, the number of engagement arms **22** is typically equal to the number of aortic sinuses **164** of the native valve, and the engagement arms are radially separated by approximately equal angles. The three-dimensional shape of engagement arms **22** causes the ends of the engagement arms to find the lowest point of reach within the floors of the sinuses, thereby enabling self-alignment of prosthesis **10** with the native aortic valve site and commissures **170**.

A length **L** (parallel to a longitudinal axis of prosthesis **10**) between (a) each juncture **24** and (b) the contact point of respective engagement arm **22** to the sinus floor is typically greater than about 6 mm, e.g., greater than about 10 mm, or than about 13 mm. For some applications, length **L** is between about 10 and about 18 mm, e.g., about 13 mm.

In typical human subjects, the native valve complex has three native commissures **170**, which define respective commissural high points, and three respective sinus low points. Prosthesis **10** is configured to match these high and low

points. Such matching enables axial anchoring, without forced bending, crimping, or folding over of the leaflets, and without impinging on the commissures. In this way, prosthesis **10** embraces the leaflets, rather than squeezing them.

For some applications, engagement arms **22** are generally aligned with the native leaflets, thereby avoiding local deformation, and distributing force over a larger contiguous area of the leaflet surface.

FIG. 8C shows valve prosthesis **10** in situ upon completion of the implantation procedure, in accordance with an embodiment of the present invention. In this embodiment, junctures **24** between pairs of engagement arms **22** ride above respective native commissures **170**, impinging on the commissures (i.e., touching or pushing the commissures). Engagement arms **22** are positioned within aortic sinuses **164**, such that the ends of the engagement arms do not reach the floors of the sinuses (such as described hereinabove with reference to FIG. 7B). The three-dimensional shape of junctures **24** causes the junctures to align with commissures **170**, thereby enabling self-alignment of prosthesis **10** with the native aortic valve site and commissures **170**. In an embodiment (not shown), junctures **24** apply axial force to (i.e., push) the commissures, and engagement arms **22** apply axial force to aortic sinuses **164**.

Reference is made to FIGS. 9A-G, which schematically illustrate a retrograde transaortic approach for implanting valve prosthesis **10**, in accordance with an embodiment of the present invention. Prior to the implantation procedure, prosthesis **10** is positioned in a retrograde delivery catheter **250**, as shown in FIG. 9G. A retrograde delivery catheter tube **251** of catheter **250** holds engagement arms **22**, and a delivery catheter cap **252** holds proximal skirt **32**.

The implantation procedure begins with the transaortic insertion of a guidewire **190** into left ventricle **157**, as shown in FIG. 9A. Optionally, stenotic aortic valve **140** is partially dilated to about 15-20 mm (e.g., about 16 mm), typically using a standard valvuloplasty balloon catheter. (In contrast, full dilation would be achieved by using a balloon catheter with a diameter of 20 mm or more.) Retrograde delivery catheter **250** is advanced over guidewire **190** into ascending aorta **160** towards native aortic valve **140**, as shown in FIG. 9A. As shown in FIG. 9B, retrograde delivery catheter **250** is advanced over guidewire **190** until delivery catheter cap **252** passes through native aortic valve **140** partially into left ventricle **157**. As also shown in FIG. 9B, retrograde delivery catheter tube **251** is pulled back (in the direction indicated by an arrow **255**), while a device stopper **254** (shown in FIG. 9G) prevents valve prosthesis **10** within tube **251** from being pulled back with tube **251**, so that engagement arms **22** are released and flare out laterally into the sinuses. At this stage of the implantation procedure, proximal skirt **32** of prosthesis **10** remains in delivery catheter cap **252**.

As shown in FIG. 9C, at the next step of the implantation procedure, delivery catheter cap **252** is pushed in the direction of the apex of the heart (as shown by an arrow **257**), using a retrograde delivery catheter cap shaft **253** that passes through tube **251** and prosthesis **10**. This advancing of cap **252** frees proximal skirt **32** to snap or spring open, and engage the inner surface of LVOT **180**. Barbs **120**, if provided, pierce or protrude into the aortic annulus on the left-ventricular side of the native valve. Retrograde delivery catheter tube **251** is further pulled back until the rest of valve prosthesis **10** is released from the tube, as shown in FIG. 9D.

Retrograde delivery catheter tube **251** is again advanced over shaft **253** toward the apex of the heart, until tube **251** rejoins cap **252**, as shown in FIG. 9E. Retrograde delivery catheter **250** and guidewire **190** are withdrawn from left

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ventricle **157**, and then from ascending aorta **160**, leaving prosthesis **10** in place, as shown in FIG. **9F**.

FIGS. **10A** and **10B** show valve prosthesis **10** in open (systolic) and closed (diastolic) positions, respectively, in accordance with an embodiment of the present invention. For clarity of illustration, the surrounding anatomy is not shown in the figure. Collapsible pliant material **105** of valve **104** opens during systole and closes during diastole, because of the fluid forces applied thereto by the blood flow and the pressure difference between the left ventricle and the aorta. Alternatively, valve **104** comprises one or more rigid components, such as rigid leaflets, for example as described in U.S. Pat. No. 6,312,465 to Griffin et al. or U.S. Pat. No. 5,908,451 to Yeo, both of which are incorporated herein by reference. Although prosthesis **10**, including valve **104**, is shown in the figures as defining a single flow field there-through, for some applications the prosthesis and valve are configured so as to define a plurality of flow fields there-through, such as shown in several figures of the '451 patent to Yeo (e.g., FIGS. **1-3** thereof).

Reference is made to FIGS. **11A-D**, which illustrate several configurations for axially coupling valve prosthesis **10** to aortic annulus **182**, in accordance with respective embodiments of the present invention. For clarity of illustration, these figures show a spread view of the native valve, viewed from a central axis of the native valve, with native aortic valve leaflets **158** cut longitudinally and pulled to the sides.

In the configuration shown in FIG. **11A**, proximal skirt **32** of valve prosthesis **10** is shaped so as to define a single barb **120** for each leaflet **158**, such that the barbs are generally centered with respect to the leaflets and engagement arms **22**. In the configuration shown in FIG. **11B**, the proximal skirt is shaped so as to define a pair of barbs **120** for each leaflet **158**.

In the configuration shown in FIG. **11C**, each engagement arm **22** comprises at least one proximal spike **192**, which typically protrudes from a most proximal region of the engagement arm (i.e., the portion of the engagement arm closest to the apex of the heart). Spikes **192** penetrate aortic annulus **182** from the aortic side, until the spikes exit the annulus on the left-ventricular side, and engage respective barbs **120** on the left-ventricular side.

In the configuration shown in FIG. **11D**, barbs **120** penetrate aortic annulus **182** from the left-ventricular side thereof, until the barbs exit the annulus on the aortic side, and are coupled to respective engagement arms **22** in respective sinuses. For example, the ends of the barbs may be shaped as hooks, in order to hook around proximal regions of engagement arms **22**.

Reference is made to FIGS. **12A-G**, which illustrate a holding device **200** for holding valve prosthesis **10** prior to the implantation of the prosthesis, in accordance with an embodiment of the present invention. Valve prosthesis **10** is loaded into delivery tube **202** from holding device **200**, as is described hereinbelow with reference to FIGS. **13A-D**. During an implantation procedure, delivery tube **202** is advanced into an overtube or trocar, such as overtube or trocar **150**, described hereinabove with reference to FIGS. **5A-C**.

FIGS. **12A** and **12B** illustrate outer and sectional views, respectively, of holding device **200**, in accordance with an embodiment of the present invention. For some applications, holding device **200** is shaped so as to define a conical portion **204** and a tubular portion **206**. Holding device **200** comprises, for example, plastic.

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FIG. **12C** shows valve prosthesis **10** loaded in holding device **200**, in accordance with an embodiment of the present invention. The proximal end of valve prosthesis **10** is typically fully compressed within tubular portion **206**, while collapsible pliant material **105** is in at least a partially open position within conical portion **204**, so as not to deform the typically delicate material of the valve. The proximal end of the prosthesis is optionally coupled to a device holder **208**.

FIGS. **12D** and **12E** show a configuration of device holder **208**, in accordance with an embodiment of the present invention. In this configuration, device holder **208** is shaped so as to define one or more female coupling openings **209**, to which corresponding male coupling members **218** of valve prosthesis **10** are releasably coupled. For example, proximal portion **34** of proximal skirt **32** (FIGS. **1** and **2B**) may be shaped so as to define male coupling members **218**. (For clarity of illustration, proximal skirt **32** is not shown in FIG. **12E**.) For some applications, the genders of the coupling elements are reversed.

FIG. **12F** illustrates holding device **200** in storage in a jar **210** containing a preservation fluid **212** such as glutaraldehyde solution. For some applications, holding device **200** is held upright by a holder **214**. The contents of the holding device **200** are typically kept in preservation fluid **212** at all times, and jar **210** is sealed by a cover **216**.

FIG. **12G** illustrates the removal of holding device **200** from storage jar **210** prior to loading valve prosthesis **10** into delivery tube **202**, in accordance with an embodiment of the present invention. Holding device **200** and its contents are typically washed prior to loading.

Reference is now made to FIGS. **13A-D**, which illustrate the loading of valve prosthesis **10** into delivery tube **202** from holding device **200**, in accordance with an embodiment of the present invention. As shown in FIG. **13A**, a distal end of a central delivery shaft **222** includes a device holder connector **220**. Device holder connector **220** is removably coupled to device holder **208**, which is coupled (e.g., fixed) to valve prosthesis **10**. For example, device holder connector **220** and device holder **208** may comprise mating, screw-threaded male and female connectors.

As shown in FIG. **13B**, retraction, to the right in the figure, of central delivery shaft **222** pulls valve prosthesis **10**, which is at least partially compressed, into delivery tube **202**. As shown in FIG. **13C**, valve prosthesis **10** is pulled into delivery tube **202**. Valve prosthesis **10** is placed in delivery tube **202** such that engagement arms **22** extend from delivery tube **202**, and thus are free to flare outwards radially, as shown in FIG. **13D**. (The engagement arms are constrained from flaring outwards during the initial steps of an implantation procedure by an overtube or trocar into which delivery tube **202** is inserted, such as overtube or trocar **150**, described hereinabove with reference to FIGS. **5A-C**.)

Although valve prosthesis **10** has been generally described herein as being implantable in an aortic valve, in some embodiments of the present invention the valve prosthesis is configured to be placed in another cardiac valve, such as a mitral valve, tricuspid valve, or pulmonary valve (such as described hereinbelow with reference to FIG. **14**), or in a venous valve. As used herein, including in the claims, "proximal" and "upstream" mean the side of the native or prosthetic valve closer to incoming blood flow, and "distal" and "downstream" mean the side of the native or prosthetic valve closer to outgoing blood flow.

Reference is made to FIG. **14**, which is a schematic illustration of a fully-assembled valve prosthesis **300** placed

in a pulmonary valve **310**, in accordance with an embodiment of the present invention. Valve prosthesis **300** is generally similar to valve prosthesis **10**, described herein with reference to FIGS. 1-13D and 16A-17, with appropriate modifications, such as size, for placement in pulmonary valve **310**. Valve prosthesis **300** comprises two portions that are configured to axially sandwich the native pulmonary valve complex from right-ventricular **312** and pulmonary trunk **314** sides thereof.

Reference is made to FIG. 15, which is a schematic anatomical illustration showing the location of a native valve complex, in accordance with an embodiment of the present invention. As used herein, including in the claims, the "native valve complex" includes the area demarcated by a box **320**, which includes native aortic valve leaflets **158**, native valve annulus **182**, subvalvular tissue **322** on the left-ventricular side, and the lower half of the aortic sinuses **164** (i.e., up to the top of box **320**).

Reference is made to FIGS. 16A-H, which schematically illustrate another transapical technique for implanting valve prosthesis **10** (in addition to the transapical approach described hereinabove with reference to FIGS. 5A-8A), in accordance with an embodiment of the present invention. Prior to the implantation procedure, prosthesis **10** is positioned in a transapical delivery catheter **350**, as shown in FIG. 16H. A transapical delivery tube **351** of catheter **350** holds proximal skirt **32**, and a transapical delivery cap **352** holds the distal end of the valve.

The implantation procedure begins with insertion of catheter **350** through an apex of the heart, into left ventricle **157**. For example, the apex may be punctured using a standard Seldinger technique. A guidewire **390** is advanced through catheter **350** into ascending aorta **160**, as shown in FIG. 16A. Optionally, aortic valve **140** is partially dilated to about 15-20 mm (e.g., about 16 mm), typically using a standard valvuloplasty balloon catheter.

Catheter **350** is advanced over guidewire **390** through native aortic valve **140**, into ascending aorta **160**. Delivery cap **352** is advanced further into the ascending aorta, by pushing with delivery cap shaft **353**. The advancement of the delivery cap releases engagement arms **22**, which flare out laterally, as shown in FIG. 16B. Catheter **350** is withdrawn towards the ventricle, thereby positioning engagement arms **22** in the sinuses, as shown in FIG. 16C. (Although engagement arms **22** are shown in FIG. 16C as being in contact with the sinus floors, for some applications the engagement arms do not come in contact with the sinus floors, such as described hereinabove with reference to FIG. 7B.) At this stage of the implantation procedure, proximal skirt **32** remains in tube **351**.

Alternatively, catheter **350** is placed within an overtube (not shown), similar to overtube or trocar **150** (FIGS. 5A-6B), and in such a configuration the engagement arms may be released either by pulling back of the overtube, or by the pushing forward of delivery end cap **352**.

At the next step of the implantation procedure, tube **351** is withdrawn in the direction of the apex of the heart. Delivery cap shaft **353** prevents cap **352** from being withdrawn with tube **351** (FIG. 16H). As a result, proximal skirt **32** is freed from tube **351** to snap or spring open, and engage the inner surface of LVOT **180**. Barbs **120**, if provided, pierce or protrude into the aortic annulus on the left-ventricular side of the native valve. It is noted that cap **352** remains in place until after proximal skirt **32** opens. Blood flow thus cannot wash the skirt downstream during the implantation procedure.

Cap **352** is advanced further into the ascending aorta by pushing on delivery cap shaft **353**, thereby releasing the rest of valve prosthesis **10** from cap **352**, as shown in FIG. 16E. Delivery tube **351** is advanced over shaft **353** through aortic valve **140**, until tube **351** rejoins cap **352**, as shown in FIG. 16F. Delivery catheter **350** is withdrawn into the left ventricle, as shown in FIG. 16G, and then from the heart, along with guidewire **390**. Prosthesis **10** is left in place, completing the implantation procedure.

Reference is made to FIG. 17, which is a schematic illustration showing a shape of engagement arms **22**, in accordance with an embodiment of the present invention. In the figure, outer support structure **14** is shown placed on an abstract geometric form **400** for clarity of illustration of the shape of the structure. As can be seen, in this embodiment engagement arms **22** have a shape that is generally upwardly concave (except at the junctures), i.e., concave in a downstream direction. In mathematical terms, this shape can be characterized by the function $z''(r) > 0$, where z is the height at any given point on one of engagement arms **22** (e.g., point P), and r is the distance from the z -axis to the given point. (It is understood that the arms may be shaped so as to include one or more relatively short sections that are upwardly convex (i.e., $z''(r) < 0$), but that the general shape of the arms is upwardly concave.)

For some applications, engagement arms **22** are shaped such that at least a portion of the arms is parallel to the longitudinal axis of outer support structure **14**.

In an embodiment, the shape of the arms is characterized by the function $z''(r) \leq 0$, i.e., the general shapes of the arms is not upwardly concave.

As used herein, including in the claims, the "ascending aorta" includes the aortic root (sinuses) and the tubular portion above the root.

Although valve prostheses **10** and **300** have been described herein as comprising a valve, for some applications the prostheses do not comprise valves.

FIG. 18 is a schematic illustration of an assembled valve prosthesis support structure **1800**, in accordance with an alternative embodiment presented herein. Assembled valve prosthesis support structure **1800** comprises a collapsible inner support structure **1812**, which serves as a proximal fixation member, and a collapsible outer support structure **1814**, which serves as a distal fixation member.

FIG. 19A is a schematic illustration of collapsible outer support structure **1814**. FIG. 19B is a schematic illustration of collapsible inner support structure **1812**. Outer and inner support structures **1814** and **1812** may be initially formed separately and then joined together, as shown in FIG. 18, or may be formed as one integrated structure. In one embodiment, outer and inner support structures **1814** and **1812** are joined together prior to implantation of the prosthesis (e.g., during a manufacturing process, or by a healthcare worker prior to implantation). In another embodiment, the outer and inner support structures **1814** and **1812** are coupled to one another during an implantation procedure. In one embodiment, outer support structure **1814** is constructed from a plurality of separate pieces, which are joined to inner support structure **1812** using standard manufacturing means, such as welding, gluing, or suturing, such that the functionality of outer support structure **1814** is attained.

Assembled valve prosthesis support structure **1800** is configured to include a valve prosthesis as described above with respect to FIG. 1. The support structure **1800** and valve prosthesis is configured to be placed in a native diseased valve of a subject, such as a native stenotic aortic or pulmonary valve, using a minimally-invasive approach,

such as a beating heart transapical procedure, such as described with reference to FIGS. 5A-8A, or with reference to FIGS. 16A-H, or a retrograde transaortic procedure, such as described with reference to FIGS. 9A-G.

As shown in FIG. 19A, outer support structure **1814** is shaped so as to define a plurality of distal strut supports **1820**, from which a plurality of proximal engagement arms **1822** extend radially outward in a proximal direction. Strut supports **1820** and engagement arms **1822** may be formed as one integrated structure (as shown), or, alternatively, may be initially formed separately and then joined to one another. For example, the strut supports **1820** and engagement arms **1822** may be mechanically interlocked, sutured, or coupled by other means. Typically, the strut support **1820** and engagement arms **1822** are joined prior to implantation.

Typically, engagement arms **1822** have a shape that is generally upwardly concave, such as described with reference to FIG. 17. Engagement arms **1822** are typically configured to be at least partially disposed within the aortic sinuses of the subject, to engage and/or rest against floors of the aortic sinuses, and/or to apply an axial force directed toward the left ventricle of the subject. Placement of engagement arms **1822** within the aortic sinuses prevents “device migration,” i.e., undesired retrograde (towards the left ventricle) movement of the valve prosthesis that may result from fluid forces applied to the valve. Engagement arms **1822** meet one another at respective junctures **1824**. In one embodiment, engagement arms **1822** are fabricated as separate pieces and are mechanically engaged to one another at respective junctures **1824**. In another embodiment, engagement arms **1822** meet via an area defined at each respective juncture **1824**, and thus do not actually touch one another. Typically, the engagement arms are configured to define respective peaks at junctures **1824**, and respective troughs **1826** between each two of the peaks.

Outer support structure **1814** is typically formed of a suitable material that allows mechanical deformations associated with crimping and expansion of a valve prosthesis. For example, outer support structure **1814** may be formed of nitinol or a stainless steel alloy (e.g., AISI 316). In one embodiment, outer support structure **1814** is fabricated from a single piece. In another embodiment, outer support structure **1814** is formed of a plurality of pieces that are coupled together. In one embodiment, engagement arms **1822** are coated with a flexible material (e.g., polyester, other synthetic material, pericardium, and/or other biological material).

With reference to FIG. 19B, inner support structure **1812** is shaped so as to define a plurality of distal inner strut supports **1830**. In one embodiment, inner support structure **1812** is sutured to outer support structure **1814** at respective strut supports **1820** and **1830**. Sutures between inner and outer support structures **1812** and **1814** allow for movement and compliance between the inner and outer support structures. Inner support structure **1812** also includes a proximal skirt **1832** that extends from the struts. A proximal portion **1834** of proximal skirt **1832** is provided in a bulging configuration to engage the left ventricular outflow tract (LVOT) of the subject and/or periannular tissue at the top of the left ventricle. A relatively narrow throat portion **1836** of proximal skirt **1832** is configured to be positioned at a valvular annulus of the subject. A flared distal portion **1837** of proximal skirt **1832** is configured to flare outward so as to engage the inner surface of the native valve leaflets. The flared distal portion **1837** preserves prosthesis leaflet integrity and prolongs overall valve durability by avoiding contact between the prosthesis leaflets and inner support struc-

ture **1812** when the prosthesis leaflets are open. Inner support structure **1812** may be formed of materials such as: nitinol, stainless steel alloys, other metals or metal alloys, or biocompatible materials.

In the embodiment shown in FIG. 18, support structure **1800** has been designed to provide a more accommodating fit within bulkier, calcified native valve commissures. For example, a distance **1841** is created between engagement arms **1822** of outer support structure **1814**. In one embodiment, distance **1841** is between about 0.5 millimeters (mm) to about 8 mm; alternatively between about 1 mm to about 6 mm, or between about 3 mm and about 5 mm. Distance **1841** allows for the accommodation of bulkier, calcified commissures, and allows support structure **1800** to sit lower in the aortic root even in the face of moderate-heavy calcification. A distance **1843** is also created between each respective trough **1826** of each engagement arm **1822** and skirt portion **1832** of inner support structure **1812**. Distance **1843** allows for the accommodation of bulkier, calcified native valve leaflets. In one embodiment, distance **1843** is between about 1 mm to about 14 mm; alternatively between about 2 mm to about 10 mm, or between about 3 mm to about 8 mm, or between about 4 mm to about 6 mm. More specifically, depending on the size of the prosthesis, distance **1843** is between about 1 mm and about 10 mm; alternatively between about 1 mm to about 12 mm, or between about 1 mm to about 14 mm; alternatively, between about 2 mm and about 9 mm, or between about 2 mm to about 10 mm; alternatively between about 3 mm to about 8 mm; alternatively between about 4 mm to about 6 mm. The design of support structure **1800** provides a frame that can conform to the patient-specific anatomy, including variability in calcium loads, while maintaining the correct position of the support structure, and therefore ensuring proper valve function. The design of support structure **1800** also ensures stable implantation and localization of the valve prosthesis, and proper sealing between the prosthesis and the native valve against perivalvular leaks.

The scope of the present invention includes embodiments described in the following applications, which are assigned to the assignee of the present application and are incorporated herein by reference. In an embodiment, techniques and apparatus described in one or more of the following applications are combined with techniques and apparatus described herein:

U.S. patent application Ser. No. 11/024,908, filed Dec. 30, 2004, entitled, “Fluid flow prosthetic device,” which published as US Patent Application Publication 2006/0149360; International Patent Application PCT/IL2005/001399, filed Dec. 29, 2005, entitled, “Fluid flow prosthetic device,” which published as PCT Publication WO 06/070372; and/or International Patent Application PCT/IL2004/000601, filed Jul. 6, 2004, entitled, “Implantable prosthetic devices particularly for transarterial delivery in the treatment of aortic stenosis, and methods of implanting such devices,” which published as PCT Publication WO 05/002466, and U.S. patent application Ser. No. 10/563,384, filed Apr. 20, 2006, in the national stage thereof, which published as US Patent Application Publication 2006/0259134.

It will be appreciated by persons skilled in the art that the present invention is not limited to what has been particularly shown and described hereinabove. Rather, the scope of the present invention includes both combinations and subcombinations of the various features described hereinabove, as well as variations and modifications thereof that are not in the prior art, which would occur to persons skilled in the art upon reading the foregoing description.

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What is claimed is:

1. A valve prosthesis comprising:

a self-expanding support comprising:

exactly three engagement arms that meet one another at three respective junctures, wherein the engagement arms are shaped so as to define three troughs;

three peak complexes at the three respective junctures, each trough disposed between two of the peak complexes and each peak complex comprising three local peaks and two local troughs;

exactly three support arms;

exactly three commissure posts, each commissure post disposed at a corresponding one of the three respective junctures;

wherein a first engagement arm of the exactly three engagement arms includes a first end coupled to a first juncture of the three respective junctures and a second end coupled to a second juncture of the three respective junctures, and a first trough of the three troughs is disposed between the first end and the second end of the first engagement arm,

wherein a second engagement arm of the exactly three engagement arms includes a first end coupled to the second juncture and a second end coupled to a third juncture of the three respective junctures, and a second trough of the three troughs is disposed between the first end and the second end of the second engagement arm,

wherein a third engagement arm of the exactly three engagement arms includes a first end coupled to the third juncture and a second end coupled to the first juncture, and a third trough of the three troughs is disposed between the first end and the second end of the third engagement arm,

wherein a first support arm of the exactly three support arms includes a first end coupled to the first engagement arm between the first juncture and the first trough and a second end coupled to the first engagement arm between the first trough and the second juncture,

wherein a second support arm of the exactly three support arms includes a first end coupled to the second engagement arm between the second juncture and the second trough and a second end coupled to the second engagement arm between the second trough and the third juncture, and

wherein a third support arm of the exactly three support arms includes a first end coupled to the third engagement arm between the third juncture and the third trough and a second end coupled to the third engagement arm between the third trough and the first juncture;

an inflow frame portion including a row of cells, wherein the row of cells defines an inflow end of the inflow frame portion and the inflow frame portion is configured to engage a native annulus of a native valve;

a graft covering coupled to the inflow frame portion; three coverings, each covering coupled to a corresponding trough of the three troughs; and

a prosthetic valve coupled to the support;

wherein the valve prosthesis is configured to be disposed in a native valve complex having three semilunar sinuses and three native commissures, wherein each of the exactly three engagement arms is configured to be at least partially disposed within a respective one of the

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three semilunar sinuses such that each trough is disposed within the respective one of the three semilunar sinuses.

2. The valve prosthesis of claim 1, wherein the prosthetic valve comprises exactly three leaflets, wherein adjacent ends of adjacent leaflets form commissures that are coupled to a corresponding one of the exactly three commissure posts and each commissure is sutured to the corresponding commissure post.

3. The valve prosthesis of claim 1, wherein the native valve complex is a native aortic valve.

4. The valve prosthesis of claim 1, wherein the native valve complex is a native pulmonary valve.

5. The valve prosthesis of claim 1, wherein at least one of the three peak complexes is configured to be disposed distal to and in rotational alignment with a respective one of the three native commissures.

6. The valve prosthesis of claim 1, wherein each covering is disposed in an area defined by a respective engagement arm and a respective support arm.

7. The valve prosthesis of claim 1, wherein each of the exactly three support arms is configured to apply pressure to an outer surface of a native leaflet of a native valve.

8. The valve prosthesis of claim 1, wherein the support is formed as a single integrated structure.

9. The valve prosthesis of claim 1, wherein the support is formed of separate pieces joined together.

10. The valve prosthesis of claim 1, wherein each trough includes a protrusion.

11. The valve prosthesis of claim 1, wherein the inflow frame portion comprises a plurality of rows of cells, wherein the row of cells is one of the plurality of rows of cells.

12. The valve prosthesis of claim 1, wherein each engagement arm extends towards the inflow end from the respective junctures, and wherein each support arm extends away from the inflow end from the respective engagement arm.

13. The valve prosthesis of claim 1, wherein each engagement arm is generally concave relative to an outflow end of the valve prosthesis and each support arm is generally concave relative to an inflow end of the valve prosthesis.

14. The valve prosthesis of claim 1, wherein each cell of the row of cells is generally diamond-shaped.

15. The valve prosthesis of claim 1, wherein each cell of the row of cells is formed from exactly four struts.

16. The valve prosthesis of claim 1, wherein each peak complex comprises exactly three local peaks and exactly two local troughs.

17. The valve prosthesis of claim 1, wherein each of the exactly three commissure posts defines holes receiving sutures therethrough.

18. The valve prosthesis of claim 17, wherein the sutures couple the commissure posts to a pliant material of the prosthetic valve.

19. The valve prosthesis of claim 18, further comprising a covering coupled to the pliant material, wherein the sutures couple the commissure posts to the covering.

20. The valve prosthesis of claim 17, wherein each of the exactly three commissure posts further defines a slit.

21. The valve prosthesis of claim 20, wherein each slit couples the commissure posts to a pliant material of the prosthetic valve.

22. The valve prosthesis of claim 21, further comprising a covering coupled to the pliant material, wherein the sutures couple the commissure posts to the covering.

23. The valve prosthesis of claim 20, wherein each slit has rounded edges.

- 24.** A method of deploying the valve prosthesis of claim 1, comprising:
- delivering the valve prosthesis disposed in a delivery catheter to a native valve;
 - proximally retracting a delivery catheter tube of the delivery catheter such that the exactly three engagement arms are released from the delivery catheter tube and flare outwardly while the inflow frame portion remains within a distal portion of the delivery catheter and the exactly three commissure posts are not released;
 - aligning the exactly three engagement arms with a respective sinus of the native valve;
 - distally advancing the distal portion of the delivery catheter such that the inflow frame portion is released from the distal portion of the delivery catheter; and
 - releasing the commissure posts after releasing the inflow frame portion.
- 25.** The method of claim **24**, further comprising:
- determining a correct rotational disposition of the valve prosthesis with respect to the sinuses of the native valve; and
 - rotating the valve prosthesis to the correct rotational disposition.
- 26.** The method of claim **25**, wherein determining the correct rotational disposition is based on tactile feedback.
- 27.** The method of claim **25**, further comprising:
- confirming the correct rotational disposition of the valve prosthesis using imaging techniques.

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